

# Root-Label Fields on Pratt Forests

An Application-First Multivariate Theory of Complete Fronts, Coordinate Transport, Arithmetic Fibres, and Monodromy

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Draft prepared with mathematical assistance from ChatGPT 5.6

Version 0.3 — computationally audited application-first edition — August 6, 2026

**Status and physical-language disclaimer.** This article is a mathematical research note. Terms such as *field*, *partition function*, *energy*, *temperature*, *Hamiltonian*, *state*, and *susceptibility* are used as structural analogies borrowed from statistical mechanics and quantum information. No claim is made that the constructions describe a real physical system, a fundamental law of nature, or an experimentally measurable phenomenon. Whenever a physical term is used, its exact mathematical definition is given.

## Abstract

This paper is organised around four concrete computations rather than around definitions alone. First, a single root-label master polynomial

$$\mathcal{M}_n((z_{p,q}))$$

recovers the exact distribution of complete fronts of the Pratt forest of an integer  $n$ , including expectations, covariances, and canonical Gram matrices of selected prime labels. Second, the Pratt incidence matrix

$$C_{q,p} = m_q(p)$$

turns ordinary prime-factor data into complete recursive Pratt coordinates by the linear rule  $m(n) = Cv(n)$ , so that large families of forests can be analysed without redrawing their trees. Third, the root-evaluation specialisation

$$F_n((x_p)_p) = \prod_p f_p(x_p)^{v_p(n)}$$

contains every completely multiplicative function as an algebraic fibre. In particular, choosing  $f_p(\alpha_p) = -1$  gives the Liouville function, and the prime number theorem becomes a cancellation statement for the partial-sum polynomials at that fibre. Fourth, the factorisation of  $p-1$  and the first row of  $C$  determine explicit ramification data for the cover  $f_p(X) - T$ ; discriminants and functional decompositions then give effective tests for symmetric or imprimitive monodromy.

The formal theory unifies two previous viewpoints. View A gives a separate variable to every root-prime type, while View B marks every prime label selected by a complete node front. The variable  $z_{p,q}$  remembers both the component root  $p$  and the selected internal label  $q$ . We define all terms from the beginning and prove the stated identities. Abstract inequalities for finite rooted forests are specialised to Pratt forests, yielding

bounds for leaf counts, front counts, polynomial values, heights, and entries of  $C$ . Positive parameters define finite exponential families whose logarithmic derivatives are rigorous probabilistic and information-geometric invariants.

Terms such as field, energy, partition function, state, susceptibility, density matrix, and modular Hamiltonian are used only as mathematical analogies. No claim is made that the constructions describe a physically real system.

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**C References and source status****60**

## 1 Four concrete tasks and the reader's payoff

The purpose of the root-label construction is not to introduce variables for their own sake. It provides a common computational language for four problems that otherwise appear separate.

| Problem              | Input                                                            | Output                                                                                                                  |
|----------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Front statistics     | the factorisation of $n$ and the component polynomials $\Phi_p$  | exact probabilities, means, covariances, and conservation laws for complete fronts                                      |
| Coordinate transport | the ordinary valuation vector $\mathbf{v}(n)$ and the matrix $C$ | the complete Pratt vector $m(n) = C\mathbf{v}(n)$ , and conversely $\mathbf{v}(n) = C^{-1}m(n)$                         |
| Arithmetic fibres    | target values $a(p)$ assigned to primes                          | a point $(\beta_p)_p$ with $F_n((\beta_p)) = a(n)$ for every completely multiplicative $a$                              |
| Monodromy tests      | $p - 1$ , $f_p$ , and $C_{2,q} = m_2(q)$                         | a forced branch cycle over $T = 1$ , a discriminant test for Morse monodromy, and decomposition tests for imprimitivity |

The central polynomial is

$$\mathcal{M}_n((z_{p,q})),$$

where the first index  $p$  records the root type of a component and the second index  $q$  records the prime label of a selected vertex. Two marginal views are recovered:

$$\text{View A: } z_{p,2} = x_p, \quad z_{p,q} = 1 \quad (q > 2),$$

$$\text{View B: } z_{p,q} = y_q \quad \text{for all } p.$$

View A preserves the component of origin and observes only terminal label 2. View B preserves every selected label but forgets the component of origin. The master polynomial retains both.

The paper therefore follows an application-first order. A complete numerical example is given before the formal definitions. After the definitions, four application sections show exactly what can be computed. The later sections then develop inequalities, field transformations, probability geometry, and further consequences.

## 2 A worked example before the formal theory: the integer 30

This section previews the entire construction with one small integer. Formal definitions and proofs begin in the next section.

The factorisation

$$30 = 2 \cdot 3 \cdot 5$$

produces three Pratt-tree components. The component  $T_2$  is a single vertex labelled 2. The component  $T_3$  has a root labelled 3 and one child labelled 2. The component  $T_5$  has a root labelled 5 and two children labelled 2, because  $5 - 1 = 2^2$ .

A complete front must choose the unique vertex in  $T_2$ . In  $T_3$  it chooses either the root 3 or the child 2. In  $T_5$  it chooses either the root 5 or both terminal 2-vertices. Hence there are  $1 \cdot 2 \cdot 2 = 4$  complete fronts.

A selected vertex with label  $q$  inside a component rooted at  $p$  receives weight  $z_{p,q}$ . The master polynomial is therefore

$$\mathcal{M}_{30} = z_{2,2}(z_{3,3} + z_{3,2})(z_{5,5} + z_{5,2}^2).$$

This single formula already gives four useful outputs.

**Output 1: View A.** Set  $z_{p,2} = x_p$  and set all other variables equal to 1. Then

$$F_{30}(x_2, x_3, x_5) = x_2(1 + x_3)(1 + x_5^2).$$

The three factors remain separated by their root types.

**Output 2: View B.** Set  $z_{p,q} = y_q$ , independently of  $p$ . Then

$$\Phi_{30}(\mathbf{y}) = y_2(y_3 + y_2)(y_5 + y_2^2).$$

The root origins are forgotten, but all selected labels remain visible.

**Output 3: the original polynomial and an exact probability law.** Set  $y_2 = x$  and  $y_q = 1$  for  $q > 2$ . This gives

$$f_{30}(x) = x(1 + x)(1 + x^2) = x + x^2 + x^3 + x^4.$$

Thus a uniformly chosen complete front selects 1, 2, 3, or 4 terminal 2-vertices, each with probability 1/4. Consequently

$$\mathbb{E}[X_{30}] = \frac{5}{2}, \quad \text{Var}(X_{30}) = \frac{5}{4}.$$

**Output 4: a covariance Gram matrix.** Let  $N_2, N_3, N_5$  count the selected labels in a uniformly random complete front. Directly from the four choices,

$$\mathbb{E}(N_2, N_3, N_5) = \left( \frac{5}{2}, \frac{1}{2}, \frac{1}{2} \right),$$

and

$$\text{Cov}(N_2, N_3, N_5) = \begin{pmatrix} \frac{5}{4} & -\frac{1}{4} & -\frac{1}{2} \\ -\frac{1}{4} & \frac{1}{4} & 0 \\ -\frac{1}{2} & 0 & \frac{1}{4} \end{pmatrix}.$$

The matrix is positive semidefinite because it is a covariance matrix. It is singular for a transparent combinatorial reason: every complete front satisfies the exact conservation law

$$N_2 + N_3 + 2N_5 = 4.$$

The weights 1, 1, 2 are precisely  $m_2(2), m_2(3), m_2(5)$ .

Finally, the ordinary prime-factor vector of 30 has one root of each type 2, 3, 5. The corresponding Pratt-coordinate columns are

$$C_{\bullet,2} = (1, 0, 0)^T, \quad C_{\bullet,3} = (1, 1, 0)^T, \quad C_{\bullet,5} = (2, 0, 1)^T.$$

Their sum is

$$m(30) = (4, 1, 1)^T.$$

Thus the same example shows the two main roles of the theory:  $\mathcal{M}_n$  analyses random complete fronts, while  $C$  analyses the deterministic population of the entire forest.

### 3 Computational verification protocol

Every theorem, proposition, lemma, and corollary in this version is accompanied by a finite computational check identifier V01–V73. The checks were run with a reproducible Python program supplied with the source archive. The program uses Python’s exact integer arithmetic together with `SymPy` for symbolic polynomials, discriminants, resultants, decompositions, rational matrices, and small Galois-group computations; `NetworkX` for exhaustive generation of nonisomorphic rooted trees; `NumPy` and `SciPy` for covariance spectra, finite differences, and stable log-partition calculations; and `mpmath` for high-precision numerical root evaluations.

The principal finite ranges are

$$n \leq 500, \quad p \leq 127, \quad n \leq 60 \text{ for exhaustive Pratt-front enumeration,}$$

and all unlabeled rooted forests with at most seven vertices are tested exhaustively. Several symbolic multivariate checks use smaller bounds because the number of monomials grows rapidly. The exact scope of each check is listed in Appendix A.

| Type of statement                        | Computational treatment                                                                                                                                                                     |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite algebraic identity                | Exact symbolic or integer equality; no floating-point tolerance.                                                                                                                            |
| Tree or front theorem                    | Exhaustive enumeration in the stated finite range.                                                                                                                                          |
| Inequality                               | Exact evaluation on all tested objects and, where useful, boundary families.                                                                                                                |
| Probability or Gram statement            | Direct front enumeration, exact moments when possible, and numerical eigenvalue or finite-difference diagnostics.                                                                           |
| Asymptotic or classical analytic theorem | Finite diagnostic only, explicitly labelled as such; no finite experiment is presented as a proof.                                                                                          |
| Generic monodromy theorem                | Exact discriminant and decomposition tests together with small rational specialisations; these are evidence and consistency checks, not a computation of the generic group in every degree. |

All 73 checks passed in the recorded run. This statement has a deliberately limited meaning: the code found no counterexample in the stated finite domains and reproduced the symbolic identities that were assigned to exact computation. Universal theorems remain justified by their written proofs. Statements involving limits, density, the prime number theorem, or generic monodromy cannot be established by finite experimentation, and the verification ledger marks those cases as diagnostics.

The worked example for  $n = 30$  is also regenerated by the script: it reproduces the master polynomial, both marginal views, the four-point terminal-count distribution, the mean  $5/2$ , the variance  $5/4$ , the covariance matrix, the conservation vector, and the Pratt coordinate vector  $(4, 1, 1)^T$ .

## 4 Arithmetic preliminaries

### 4.1 Prime valuations and arithmetic functions

Let  $\mathbb{P} = \{2, 3, 5, \dots\}$  denote the set of prime numbers.

**Definition 4.1** (Prime valuation). For  $n \in \mathbb{N}$  and  $p \in \mathbb{P}$ , the number  $v_p(n) \in \mathbb{N}_0$  is the unique exponent such that  $p^{v_p(n)} \mid n$  and  $p^{v_p(n)+1} \nmid n$ .

The fundamental theorem of arithmetic gives

$$n = \prod_{p \in \mathbb{P}} p^{v_p(n)},$$

where only finitely many factors differ from 1.

**Definition 4.2** (Completely additive and completely multiplicative). A function  $a : \mathbb{N} \rightarrow \mathbb{C}$  is *completely additive* if

$$a(mn) = a(m) + a(n)$$

for all  $m, n \in \mathbb{N}$ . A function  $b : \mathbb{N} \rightarrow \mathbb{C}$  is *completely multiplicative* if

$$b(mn) = b(m)b(n)$$



for all  $m, n \in \mathbb{N}$ .

No coprimality condition is present in either definition.

**Definition 4.3** (Big Omega and Liouville function). Set

$$\Omega(n) := \sum_{p \in \mathbb{P}} v_p(n), \quad \lambda(n) := (-1)^{\Omega(n)}.$$

**Proposition 4.4.** *The function  $\Omega$  is completely additive and the Liouville function  $\lambda$  is completely multiplicative.*

**Finite computational check V01: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* For every prime  $p$ ,  $v_p(mn) = v_p(m) + v_p(n)$ . Summing over  $p$  gives complete additivity of  $\Omega$ . Therefore

$$\lambda(mn) = (-1)^{\Omega(mn)} = (-1)^{\Omega(m) + \Omega(n)} = \lambda(m)\lambda(n).$$

□

## 5 Pratt trees, Pratt forests, and Pratt coordinates

### 5.1 Pratt trees

**Definition 5.1** (Pratt tree). The Pratt tree  $T_p$  of a prime  $p$  is defined recursively.

- $T_2$  consists of one vertex, the root, labelled 2.
- If  $p > 2$  is prime, the root is labelled  $p$ . For each prime  $q \mid p-1$ , the root has  $v_q(p-1)$  children, and below each such child is a copy of  $T_q$ .

The construction terminates because every child label  $q$  satisfies  $q \leq p-1 < p$ .

**Definition 5.2** (Pratt forest). For  $n \in \mathbb{N}$ , the Pratt forest  $\mathcal{F}_n$  is the finite disjoint union

$$\mathcal{F}_n := \bigsqcup_{p \in \mathbb{P}} v_p(n) T_p,$$

where  $v_p(n) T_p$  denotes  $v_p(n)$  disjoint copies of  $T_p$ . The forest  $\mathcal{F}_1$  is empty.

**Definition 5.3** (Pratt coordinate). For primes  $q$  and an integer  $n \geq 1$ , define

$$m_q(n) := \#\{\text{vertices of } \mathcal{F}_n \text{ labelled } q\}.$$

The finitely supported vector

$$m(n) := (m_q(n))_{q \in \mathbb{P}}$$

is called the Pratt-coordinate vector of  $n$ .

**Proposition 5.4** (Complete additivity). *For every prime  $q$  and all  $a, b \in \mathbb{N}$ ,*

$$m_q(ab) = m_q(a) + m_q(b).$$

**Finite computational check V02: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The ordinary valuation identity  $v_p(ab) = v_p(a) + v_p(b)$  implies that  $\mathcal{F}_{ab}$  is, up to relabelling of disjoint copies, the disjoint union  $\mathcal{F}_a \sqcup \mathcal{F}_b$ . Counting  $q$ -labelled vertices in a disjoint union gives the asserted sum.  $\square$

## 6 The recursive polynomials $f_n(x)$

**Definition 6.1** (Pratt polynomial family). Define

$$f_1(x) := 1, \quad f_2(x) := x.$$

For an odd prime  $p$ , set

$$f_p(x) := 1 + \prod_{q|p-1} f_q(x)^{v_q(p-1)}.$$

For arbitrary  $n \geq 1$ , define

$$f_n(x) := \prod_{p \in \mathbb{P}} f_p(x)^{v_p(n)}.$$

All products are finite.

**Proposition 6.2.** *The family satisfies*

$$f_{mn}(x) = f_m(x)f_n(x)$$

for all  $m, n \in \mathbb{N}$ , and for every odd prime  $p$ ,

$$f_p(x) = 1 + f_{p-1}(x).$$

**Finite computational check V03: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The first identity follows from  $v_p(mn) = v_p(m) + v_p(n)$ . For the second,

$$f_{p-1}(x) = \prod_{q|p-1} f_q(x)^{v_q(p-1)}$$

by the multiplicative definition.  $\square$

**Theorem 6.3** (Evaluation at 2). *For every  $n \in \mathbb{N}$ ,*

$$f_n(2) = n.$$

**Finite computational check V04: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* We first prove  $f_p(2) = p$  by induction over primes. The claim is clear for  $p = 2$ . If  $p > 2$ , every prime divisor  $q$  of  $p - 1$  is smaller than  $p$ , so

$$f_p(2) = 1 + \prod_{q|p-1} f_q(2)^{v_q(p-1)} = 1 + \prod_{q|p-1} q^{v_q(p-1)} = 1 + (p - 1) = p.$$

For general  $n$ ,

$$f_n(2) = \prod_p f_p(2)^{v_p(n)} = \prod_p p^{v_p(n)} = n.$$

□

The first examples are

$$f_2 = x, \quad f_3 = 1 + x, \quad f_5 = 1 + x^2, \quad f_7 = 1 + x + x^2, \quad f_{12} = x^2(1 + x).$$

## 7 Complete node fronts

### 7.1 Rooted forests and antichains

**Definition 7.1** (Ancestor order). In a rooted tree, a vertex  $u$  is an ancestor of  $v$ , written  $u \preceq v$ , if  $u$  lies on the unique path from the root to  $v$ . Vertices in distinct components of a rooted forest are declared incomparable.

**Definition 7.2** (Node front and complete node front). A *node front* in a finite rooted forest  $F$  is an antichain for the ancestor order. It is *complete* if it meets every root-to-leaf path. The set of complete node fronts is denoted by  $\text{CF}(F)$ .

Because a root-to-leaf path is totally ordered, a node front meets such a path at most once. Thus a complete node front meets every root-to-leaf path exactly once.

**Lemma 7.3** (Maximal antichains are complete). *A node front in a finite rooted forest is complete if and only if it is maximal among node fronts under inclusion.*

**Finite computational check V05: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Suppose  $A$  is complete and  $v \notin A$ . Choose a root-to-leaf path through  $v$ . It contains a point  $a \in A$ , and  $a$  is comparable with  $v$ . Hence  $v$  cannot be added to  $A$  while preserving the antichain property, so  $A$  is maximal.

Conversely, suppose  $A$  is maximal but misses a root-to-leaf path  $P$ . Let  $\ell$  be the leaf of  $P$ . No element of  $A$  is an ancestor of  $\ell$ , since every ancestor of  $\ell$  lies on  $P$ . The leaf  $\ell$  is not an ancestor of any distinct vertex. Therefore  $\ell$  is incomparable with every element of  $A$ , contradicting maximality. Hence  $A$  is complete. □

**Corollary 7.4** (Extension lemma). *Every node front in a finite rooted forest is contained in a complete node front.*

**Finite computational check V06: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Since the vertex set is finite, enlarge the node front until it is maximal, and apply the preceding lemma. □

## 7.2 The fully label-marked front polynomial: View B

**Definition 7.5** (Selected-label counts). Let  $F$  be a prime-labelled rooted forest and let  $A \in \text{CF}(F)$ . For  $q \in \mathbb{P}$ , set

$$N_q(A) := \#\{v \in A : \ell(v) = q\},$$

where  $\ell(v)$  denotes the label of  $v$ .

**Definition 7.6** (Fully label-marked front polynomial). Define

$$\Phi_F(\mathbf{z}) := \sum_{A \in \text{CF}(F)} \prod_{q \in \mathbb{P}} z_q^{N_q(A)} = \sum_{A \in \text{CF}(F)} \prod_{v \in A} z_{\ell(v)}.$$

For a Pratt tree write  $\Phi_p := \Phi_{T_p}$ , and for a Pratt forest write  $\Phi_n := \Phi_{\mathcal{F}_n}$ .

Only finitely many variables occur in each polynomial.

**Theorem 7.7** (Front recursion). *The fully marked polynomials satisfy*

$$\Phi_2(\mathbf{z}) = z_2,$$

$$\Phi_p(\mathbf{z}) = z_p + \prod_{q|p-1} \Phi_q(\mathbf{z})^{v_q(p-1)} \quad (p > 2 \text{ prime}),$$

and

$$\Phi_n(\mathbf{z}) = \prod_p \Phi_p(\mathbf{z})^{v_p(n)}.$$

**Finite computational check V07: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* For  $T_2$  there is one complete front, namely its unique vertex. Now let  $p > 2$ . A complete front of  $T_p$  has exactly two possible forms.

If it contains the root, then it contains no descendant, and the front is the singleton root; its monomial is  $z_p$ . If it does not contain the root, then in every child subtree it must choose a complete front. The choices in distinct child subtrees are independent, so their generating polynomials multiply. There are  $v_q(p-1)$  copies of  $T_q$ , giving the displayed product.

For a disjoint union of rooted forests, a complete front is exactly a union of one complete front from each component. Consequently generating polynomials multiply. Applying this to the definition of  $\mathcal{F}_n$  proves the final identity.  $\square$

**Theorem 7.8** (Recovery of  $f_n$ ). *Under the specialisation*

$$z_2 = x, \quad z_q = 1 \quad (q > 2),$$

one has

$$\Phi_n(\mathbf{z}) = f_n(x).$$

Equivalently,

$$f_n(x) = \sum_{A \in \text{CF}(\mathcal{F}_n)} x^{N_2(A)}.$$

**Finite computational check V08: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Under the stated substitution, the recursion for  $\Phi_2$  becomes  $f_2(x) = x$ . For an odd prime, the term  $z_p$  becomes 1, while the descendant product is transformed recursively into

$$\prod_{q|p-1} f_q(x)^{v_q(p-1)}.$$

Thus the specialised  $\Phi_p$  satisfies exactly the defining recursion for  $f_p$ . Multiplicativity over forests gives the general case.  $\square$

**Remark 7.9.** The variable  $x$  in  $f_n(x)$  records only selected terminal 2-vertices. The polynomial  $\Phi_n(z)$  retains the joint count of every selected prime label.

## 8 Root-separated evaluation: View A

**Definition 8.1** (Root-evaluation polynomial). Let  $\mathbf{x} = (x_p)_{p \in \mathbb{P}}$  be a family of variables indexed by primes. Define

$$F_n(\mathbf{x}) := \prod_{p \in \mathbb{P}} f_p(x_p)^{v_p(n)}.$$

For each fixed choice of the variables,  $n \mapsto F_n(\mathbf{x})$  is completely multiplicative.

**Proposition 8.2** (Diagonal recovery). *If  $x_p = x$  for every prime  $p$ , then*

$$F_n(\mathbf{x}) = f_n(x).$$

**Finite computational check V09: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Substitution gives

$$F_n((x)_p) = \prod_p f_p(x)^{v_p(n)} = f_n(x).$$

$\square$

The variable  $x_p$  marks the terminal 2-vertices occurring inside components whose root type is  $p$ . It does not distinguish other internal labels. This is the complementary loss of information to View B: View A remembers the root type but observes only label 2, while View B remembers every label but forgets the root type from which it came.

## 9 The root-label master polynomial $\mathcal{M}_n((z_{p,q}))$

### 9.1 Definition

Each component of  $\mathcal{F}_n$  has a root label. If a vertex  $v$  lies in a component whose root is labelled  $p$ , write  $r(v) = p$ .

**Definition 9.1** (Root-label count). For  $A \in \text{CF}(\mathcal{F}_n)$  and primes  $p, q$ , define

$$N_{p,q}(A) := \#\{v \in A : r(v) = p, \ell(v) = q\}.$$

Clearly

$$N_q(A) = \sum_p N_{p,q}(A).$$

**Definition 9.2** (Master polynomial). Assign a variable  $z_{p,q}$  to every ordered pair of primes. Define

$$\mathcal{M}_n(\mathbf{z}) := \sum_{A \in \text{CF}(\mathcal{F}_n)} \prod_{p,q \in \mathbb{P}} z_{p,q}^{N_{p,q}(A)}.$$

Only pairs that actually occur in  $\mathcal{F}_n$  enter the polynomial.

For each root type  $p$ , write

$$\mathbf{z}^{(p)} = (z_{p,q})_{q \in \mathbb{P}}.$$

**Theorem 9.3** (Factorisation of the master polynomial). *For every  $n \in \mathbb{N}$ ,*

$$\mathcal{M}_n(\mathbf{z}) = \prod_{p \in \mathbb{P}} \Phi_p(\mathbf{z}^{(p)})^{v_p(n)}.$$

**Finite computational check V10: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* A complete front of  $\mathcal{F}_n$  is a union of independently chosen complete fronts, one in each component. In a component rooted at  $p$ , a selected node labelled  $q$  receives the variable  $z_{p,q}$ , so its component generating polynomial is  $\Phi_p(\mathbf{z}^{(p)})$ . There are  $v_p(n)$  components of that root type. Multiplication over all components yields the formula.  $\square$

## 9.2 Views A and B as specialisations

**Theorem 9.4** (Unification theorem). *The root-evaluation and fully label-marked constructions are specialisations of  $\mathcal{M}_n$ .*

(i) If

$$z_{p,2} = x_p, \quad z_{p,q} = 1 \quad (q > 2),$$

then

$$\mathcal{M}_n(\mathbf{z}) = F_n(\mathbf{x}).$$

(ii) If

$$z_{p,q} = y_q \quad \text{for all } p, q,$$

then

$$\mathcal{M}_n(\mathbf{z}) = \Phi_n(\mathbf{y}).$$

**Finite computational check V11: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* For (i), in every component rooted at  $p$  the substitution marks only selected 2-vertices with  $x_p$ . By the complete-front interpretation, the component polynomial becomes  $f_p(x_p)$ . The factorisation theorem gives  $F_n(\mathbf{x})$ .

For (ii), the weight of a selected node depends only on its label  $q$  and no longer on the root type. Thus

$$\prod_{p,q} z_{p,q}^{N_{p,q}(A)} = \prod_{p,q} y_q^{N_{p,q}(A)} = \prod_q y_q^{\sum_p N_{p,q}(A)} = \prod_q y_q^{N_q(A)}.$$

Summing over fronts gives  $\Phi_n(\mathbf{y})$ . □

**Corollary 9.5** (Common one-dimensional slice). *Both specialisations recover the original polynomial:*

$$F_n((x)_p) = f_n(x) = \Phi_n(y_2 = x, y_q = 1 \ (q > 2)).$$

**Finite computational check V12: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

### 9.3 Examples

**Example 9.6** (The integer 6). The forest  $\mathcal{F}_6$  consists of one  $T_2$  component and one  $T_3$  component. Since

$$\Phi_2(z_{2,\bullet}) = z_{2,2}, \quad \Phi_3(z_{3,\bullet}) = z_{3,3} + z_{3,2},$$

we obtain

$$\mathcal{M}_6 = z_{2,2}(z_{3,3} + z_{3,2}).$$

View A gives

$$F_6(x_2, x_3) = x_2(1 + x_3),$$

while View B gives

$$\Phi_6(\mathbf{y}) = y_2(y_3 + y_2).$$

Both reduce to  $f_6(x) = x(1 + x)$  on the common slice.

**Example 9.7** (The integer 12). Since  $12 = 2^2 \cdot 3$ ,

$$\mathcal{M}_{12} = z_{2,2}^2(z_{3,3} + z_{3,2}).$$

Hence

$$F_{12}(x_2, x_3) = x_2^2(1 + x_3), \quad \Phi_{12}(\mathbf{y}) = y_2^2(y_3 + y_2).$$

The diagonal specialisation is  $f_{12}(x) = x^2(1 + x)$ .

**Example 9.8** (The prime 19). Because  $19 - 1 = 2 \cdot 3^2$ ,

$$\Phi_{19}(\mathbf{y}) = y_{19} + y_2(y_3 + y_2)^2.$$

Specialising  $y_2 = x$  and  $y_q = 1$  for  $q > 2$  gives

$$f_{19}(x) = 1 + x(1 + x)^2.$$

## 10 Application I: exact front statistics from one polynomial

Once  $\mathcal{M}_n$  is known, no separate enumeration is needed to answer probabilistic questions about complete fronts.

**Theorem 10.1** (Coefficient and derivative extraction). *Choose a complete front of  $\mathcal{F}_n$  uniformly. Put*

$$Z_n := \mathcal{M}_n(\mathbf{1}) = \# \text{CF}(\mathcal{F}_n), \quad D_{p,q} := z_{p,q} \frac{\partial}{\partial z_{p,q}}.$$

Then

$$\mathbb{P}((N_{p,q}) = (a_{p,q})) = \frac{[(z_{p,q})^{(a_{p,q})}] \mathcal{M}_n}{Z_n},$$

$$\mathbb{E}[N_{p,q}] = D_{p,q} \log \mathcal{M}_n|_{z=\mathbf{1}},$$

and

$$\text{Cov}(N_{p,q}, N_{r,s}) = D_{p,q} D_{r,s} \log \mathcal{M}_n|_{z=\mathbf{1}}.$$

**Finite computational check V13: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* After division by  $Z_n$ , the polynomial  $\mathcal{M}_n$  is the multivariate probability-generating function of the count vector  $(N_{p,q})$ . Replacing  $z_{p,q}$  by  $e^{\theta_{p,q}}$  gives its moment-generating function. One derivative of the logarithm gives a mean, and two derivatives give a covariance. Coefficient extraction gives the exact mass function.  $\square$

For the one-variable leaf statistic  $X_n = N_2$ , the formulas reduce to

$$\mathbb{E}[X_n] = \frac{f'_n(1)}{f_n(1)},$$

$$\text{Var}(X_n) = \frac{f''_n(1)}{f_n(1)} + \frac{f'_n(1)}{f_n(1)} - \left( \frac{f'_n(1)}{f_n(1)} \right)^2.$$

| $n$ | $f_n(x)$                | $\# \text{CF}(\mathcal{F}_n)$ | $\mathbb{E}[X_n]$ | $\text{Var}(X_n)$ |
|-----|-------------------------|-------------------------------|-------------------|-------------------|
| 12  | $x^2 + x^3$             | 2                             | 5/2               | 1/4               |
| 19  | $1 + x + 2x^2 + x^3$    | 5                             | 8/5               | 26/25             |
| 30  | $x + x^2 + x^3 + x^4$   | 4                             | 5/2               | 5/4               |
| 60  | $x^2 + x^3 + x^4 + x^5$ | 4                             | 7/2               | 5/4               |

These examples illustrate three points. The value  $f_n(1)$  counts fronts; the coefficients give the complete distribution; and multiplication of integers becomes convolution of independent front statistics. For example,  $60 = 2 \cdot 30$  adds one deterministic selected leaf to the distribution for 30, shifting its mean by 1 without changing its variance.

The multivariate version is more informative. Negative off-diagonal covariances measure competition between choosing a root and descending to labels below it. Zero covariances identify independent component choices. Linear conservation laws produce null vectors of the covariance matrix and can therefore be detected by linear algebra.



## 11 The Pratt incidence matrix $C$

### 11.1 Definition and coordinate transport

Order the primes increasingly. Define the infinite matrix

$$C_{q,p} := m_q(p).$$

Its  $p$ -th column is the Pratt-coordinate vector of the prime tree  $T_p$ .

Let

$$\mathbf{v}(n) := (v_p(n))_{p \in \mathbb{P}}.$$

**Theorem 11.1** (Ordinary-to-Pratt coordinate transform). *For every  $n \in \mathbb{N}$ ,*

$$m(n) = C\mathbf{v}(n).$$

Equivalently,

$$m_q(n) = \sum_p C_{q,p} v_p(n).$$

**Finite computational check V14: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The forest  $\mathcal{F}_n$  contains  $v_p(n)$  copies of  $T_p$ . Each copy contributes  $m_q(p) = C_{q,p}$  vertices labelled  $q$ . Summing over root types gives the formula.  $\square$

### 11.2 The one-generation matrix and the path expansion

**Definition 11.2** (Immediate-child matrix). Define

$$R_{q,p} := \begin{cases} v_q(p-1), & p > 2, \\ 0, & p = 2. \end{cases}$$

Thus  $R_{q,p}$  counts immediate children labelled  $q$  below a root labelled  $p$ .

Because  $R_{q,p} \neq 0$  implies  $q < p$ , every finite principal truncation of  $R$  is strictly upper triangular and therefore nilpotent.

**Theorem 11.3** (Resolvent identity). *For every finite prime truncation closed under smaller primes,*

$$C = I + CR, \quad C = (I - R)^{-1}.$$

Consequently,

$$C = I + R + R^2 + \cdots + R^h$$

for some finite  $h$  depending on the truncation.

**Finite computational check V15: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Fix primes  $q, p$ . The root of  $T_p$  contributes  $\delta_{q,p}$ . Every immediate child of type  $r$  is the root of a copy of  $T_r$ , which contributes  $m_q(r) = C_{q,r}$  vertices labelled  $q$ . There are  $R_{r,p}$  such children. Therefore

$$C_{q,p} = \delta_{q,p} + \sum_r C_{q,r} R_{r,p},$$

which is  $C = I + CR$ . Rearranging gives  $C(I - R) = I$ . Since  $R$  is nilpotent on a finite truncation,  $I - R$  is invertible and its inverse is the finite geometric series.  $\square$

**Corollary 11.4** (Weighted path formula). *For primes  $q \leq p$ ,*

$$C_{q,p} = \sum_{k \geq 0} \sum_{p=p_0 > p_1 > \dots > p_k=q} \prod_{j=1}^k v_{p_j}(p_{j-1} - 1),$$

where the  $k = 0$  term is 1 if  $p = q$  and 0 otherwise.

**Finite computational check V16: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The entry  $(R^k)_{q,p}$  is the sum of products of edge multiplicities along all length- $k$  descending prime chains from  $p$  to  $q$ . Sum over  $k$  using  $C = \sum_k R^k$ .  $\square$

**Example 11.5** (A finite truncation). For the ordered primes 2, 3, 5, 7, 11, 13, the matrix is

$$C_{\leq 13} = \begin{pmatrix} 1 & 1 & 2 & 2 & 3 & 3 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

For example, the 11-column is  $(3, 0, 1, 0, 1, 0)^T$ : the tree  $T_{11}$  contains three vertices labelled 2, one labelled 5, and the root labelled 11.

### 11.3 Recovering $C$ from View B

The fully marked front polynomial does not merely encode random front statistics. Its coordinate degrees recover the deterministic node counts of the entire Pratt tree.

**Lemma 11.6** (Equal labels form an antichain). *In a Pratt forest, two distinct vertices carrying the same prime label are incomparable in the ancestor order.*

**Finite computational check V17: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Along every edge of a Pratt tree, the prime label strictly decreases. Hence labels strictly decrease along every proper descendant chain. Two vertices with the same label cannot lie on one such chain.  $\square$

**Theorem 11.7** (Degree recovery of the Pratt matrix). *For all primes  $p, q$ ,*

$$\deg_{z_q} \Phi_p(z) = C_{q,p} = m_q(p).$$

*More generally,*

$$\deg_{z_q} \Phi_n(z) = m_q(n),$$

*and*

$$\deg_{z_{p,q}} \mathcal{M}_n(z) = v_p(n) C_{q,p}.$$

**Finite computational check V18: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* No complete front can contain more  $q$ -labelled vertices than exist in the whole tree, so

$$\deg_{z_q} \Phi_p \leq C_{q,p}.$$

Let  $S$  be the set of all  $q$ -labelled vertices in  $T_p$ . By the preceding lemma,  $S$  is a node front. By the extension lemma,  $S$  is contained in a complete node front. That complete front contributes a monomial whose exponent of  $z_q$  is  $|S| = C_{q,p}$ . Thus equality holds.

For  $\Phi_n = \prod_p \Phi_p^{v_p(n)}$ , degrees add under products, yielding

$$\deg_{z_q} \Phi_n = \sum_p v_p(n) C_{q,p} = m_q(n).$$

The master-polynomial statement follows similarly from its factorisation, noting that  $z_{p,q}$  occurs only in the  $p$ -root factor.  $\square$

**Remark 11.8** (Newton-polytope interpretation). The column  $C_{\bullet,p}$  gives the coordinatewise maximal exponent vector of  $\Phi_p$ . It need not itself occur as one monomial: a complete front cannot generally select every vertex of a tree simultaneously. Thus  $C_{\bullet,p}$  is a coordinatewise capacity vector, not necessarily one point of the support.

## 11.4 A saturation limit

**Proposition 11.9** (Zero-temperature or tropical recovery). *Fix primes  $p, q$ . Set all variables of  $\Phi_p$  equal to 1 except  $z_q = e^t$ . Then*

$$\lim_{t \rightarrow +\infty} \frac{d}{dt} \log \Phi_p(z_q = e^t, z_r = 1 \ (r \neq q)) = C_{q,p}.$$

**Finite computational check V19: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Write the resulting one-variable polynomial as

$$P(e^t) = \sum_{j=0}^d a_j e^{jt}, \quad a_j \geq 0, \quad a_d > 0,$$

where  $d = \deg_{z_q} \Phi_p = C_{q,p}$ . Then

$$\frac{d}{dt} \log P(e^t) = \frac{\sum_j j a_j e^{jt}}{\sum_j a_j e^{jt}}.$$

Divide numerator and denominator by  $e^{dt}$ . Every term with  $j < d$  tends to zero, leaving  $d$ .  $\square$

In the statistical language developed later, a large positive field coupled to label  $q$  forces the mean selected  $q$ -population toward its maximal possible value  $C_{q,p}$ .

## 12 Application II: coordinate algorithms without redrawing trees

The matrix  $C$  changes the calculation of Pratt coordinates from a repeated tree-drawing problem into a triangular linear-algebra problem.

**Proposition 12.1** (Recursive construction of the columns of  $C$ ). *Let  $e_p$  denote the unit vector in the coordinate indexed by  $p$ . Then*

$$C_{\bullet,2} = e_2,$$

and for every odd prime  $p$ ,

$$C_{\bullet,p} = e_p + \sum_{q|p-1} v_q(p-1) C_{\bullet,q}.$$

**Finite computational check V20: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The root of  $T_p$  contributes  $e_p$ . Every child of type  $q$  carries an entire copy of  $T_q$ , and there are  $v_q(p-1)$  such children. Summing their population vectors gives the formula.  $\square$

Hence all columns up to a cutoff can be computed in increasing prime order. After this precomputation, an integer

$$n = \prod_p p^{v_p(n)}$$

has Pratt vector

$$m(n) = \sum_p v_p(n) C_{\bullet,p} = C \mathbf{v}(n).$$

No recursive tree expansion is required for each new integer.

Conversely, because  $C^{-1} = I - R$  on every finite truncation,

$$v_q(n) = m_q(n) - \sum_{p>q} v_q(p-1) m_p(n).$$

Thus ordinary factorisation data can be recovered from a valid Pratt vector by subtracting the descendants forced by higher labels.

**Example 12.2** (The integer 21). Using coordinates indexed by 2, 3, 7,

$$C_{\bullet,3} = (1, 1, 0)^T, \quad C_{\bullet,7} = (2, 1, 1)^T.$$

Since  $21 = 3 \cdot 7$ ,

$$m(21) = C_{\bullet,3} + C_{\bullet,7} = (3, 2, 1)^T.$$

The inverse calculation gives

$$v_7 = 1, \quad v_3 = m_3 - v_3(6)m_7 = 2 - 1 = 1,$$

$$v_2 = m_2 - v_2(2)m_3 - v_2(6)m_7 = 3 - 2 - 1 = 0.$$

Thus the ordinary root factorisation  $21 = 3 \cdot 7$  is recovered from the total recursive populations.

There is also an independent verification mechanism. The degree theorem proved above gives

$$\deg_{z_q} \Phi_n = m_q(n).$$

A computed front polynomial and a computed matrix column can therefore be checked against one another. This is useful in symbolic implementations: the tree recursion, the matrix recursion, and polynomial degrees provide three independent representations of the same data.

## 13 Application III: algebraic fibres and the prime number theorem

The benefit of View A is that it converts prescribed prime values into one algebraic point of the root-parameter space. At that point, every integer value is forced by complete multiplicativity. The Liouville function is the most important example because its summatory cancellation is equivalent to the prime number theorem.

### 13.1 The Liouville fibre

The root-evaluation family was proposed in the MathOverflow question “Are most prime numbers symmetric?” as a route connecting the recursive polynomials with Galois theory and the prime number theorem.

**Theorem 13.1** (Liouville specialisation). *For every prime  $p$ , choose a complex number  $\alpha_p$  satisfying*

$$f_p(\alpha_p) = -1.$$

*Then*

$$\boxed{F_n((\alpha_p)_p) = \lambda(n)}$$

*for every  $n \in \mathbb{N}$ .*

**Finite computational check V21: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Each  $f_p$  is nonconstant, so  $f_p(X) + 1$  has a complex root by the fundamental theorem of algebra. Then

$$F_n((\alpha_p)_p) = \prod_p f_p(\alpha_p)^{v_p(n)} = \prod_p (-1)^{v_p(n)} = (-1)^{\Omega(n)} = \lambda(n).$$

□

Define the partial-sum polynomial

$$S_N(\mathbf{x}) := \sum_{n \leq N} F_n(\mathbf{x}).$$

At the Liouville fibre,

$$S_N((\alpha_p)_p) = \sum_{n \leq N} \lambda(n) =: L(N).$$

### 13.2 Why $L(N) = o(N)$ is equivalent to the PNT

We recall a classical theorem of Landau.

**Theorem 13.2** (Landau, classical). *The prime number theorem is equivalent to*

$$M(x) := \sum_{n \leq x} \mu(n) = o(x),$$

where  $\mu$  is the Moebius function.

**Finite computational check V22: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

We now prove that  $M(x) = o(x)$  is equivalent to  $L(x) = o(x)$ .

**Lemma 13.3** (Convolution relation). *Let  $\mathbf{1}_\square(n)$  be 1 when  $n$  is a square and 0 otherwise. Then*

$$\lambda = \mu * \mathbf{1}_\square.$$

Equivalently,

$$\lambda(n) = \sum_{d^2 | n} \mu(n/d^2).$$

**Finite computational check V23: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Both sides are multiplicative, so it suffices to check  $n = p^a$ . The only nonzero term occurs when  $a - 2j$  equals 0 or 1. If  $a$  is even, the sum is  $\mu(1) = 1$ ; if  $a$  is odd, it is  $\mu(p) = -1$ . This equals  $(-1)^a = \lambda(p^a)$ . □

**Lemma 13.4.** *For  $x \geq 1$ ,*

$$L(x) = \sum_{d \leq \sqrt{x}} M(x/d^2).$$

Conversely,

$$M(x) = \sum_{d \leq \sqrt{x}} \mu(d) L(x/d^2).$$

**Finite computational check V24: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Summing the convolution relation gives

$$L(x) = \sum_{n \leq x} \sum_{d^2 | n} \mu(n/d^2) = \sum_{d \leq \sqrt{x}} \sum_{m \leq x/d^2} \mu(m) = \sum_{d \leq \sqrt{x}} M(x/d^2).$$

For the inverse identity, the Dirichlet series of the function supported on squares with value  $\mu(d)$  at  $d^2$  is  $1/\zeta(2s)$ , which is the convolution inverse of  $\mathbf{1}_{\square}$ , whose Dirichlet series is  $\zeta(2s)$ . Thus  $\mu = \lambda * h$  with  $h(d^2) = \mu(d)$ , and summation gives the formula. The same conclusion can also be checked prime-power by prime-power.  $\square$

**Proposition 13.5.** *One has*

$$M(x) = o(x) \iff L(x) = o(x).$$

**Finite computational check V25: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Assume  $M(y) = o(y)$ . Given  $\varepsilon > 0$ , choose  $Y$  such that  $|M(y)| \leq \varepsilon y$  for  $y \geq Y$ . In

$$L(x) = \sum_{d \leq \sqrt{x}} M(x/d^2),$$

split at  $d \leq \sqrt{x/Y}$ . The first part has absolute value at most

$$\varepsilon x \sum_{d \geq 1} d^{-2}.$$

For the remaining terms use  $|M(y)| \leq y$  to obtain

$$x \sum_{d > \sqrt{x/Y}} d^{-2} = O(\sqrt{xY}) = o(x).$$

Since  $\varepsilon$  is arbitrary,  $L(x) = o(x)$ .

Conversely, use

$$M(x) = \sum_{d \leq \sqrt{x}} \mu(d) L(x/d^2)$$

and the same splitting argument, with  $|\mu(d)| \leq 1$  and the trivial bound  $|L(y)| \leq y$ . This yields  $M(x) = o(x)$ .  $\square$

**Corollary 13.6** (PNT as a polynomial-fibre cancellation statement). *With the choices  $f_p(\alpha_p) = -1$ ,*

$$\boxed{\text{PNT} \iff \lim_{N \rightarrow \infty} \frac{S_N((\alpha_p)_p)}{N} = 0.}$$

**Finite computational check V26: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The specialisation equals  $L(N)$ , and the preceding proposition together with Landau's theorem gives the equivalence.  $\square$

This reformulation does not prove the prime number theorem anew. It embeds its cancellation statement into a multivariate algebraic family.

### 13.3 Euler products

**Theorem 13.7** (Universal Euler product). *Let  $s \in \mathbb{C}$ . Assume*

$$\sum_p \frac{|f_p(x_p)|}{p^{\Re s}} < \infty \quad \text{and} \quad |f_p(x_p)| < p^{\Re s} \quad \text{for every prime } p.$$

*Then*

$$\boxed{\sum_{n \geq 1} \frac{F_n(\mathbf{x})}{n^s} = \prod_p \left(1 - \frac{f_p(x_p)}{p^s}\right)^{-1}},$$

*and both sides converge absolutely.*

**Finite computational check V27: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The function  $n \mapsto F_n(\mathbf{x})$  is completely multiplicative. The second assumption makes every local geometric series convergent:

$$\sum_{k \geq 0} \frac{F_{p^k}(\mathbf{x})}{p^{ks}} = \sum_{k \geq 0} \left(\frac{f_p(x_p)}{p^s}\right)^k = \left(1 - \frac{f_p(x_p)}{p^s}\right)^{-1}.$$

The first assumption implies absolute convergence of the infinite product and of the corresponding multiplicative Dirichlet series, so the prime-power factors may be multiplied and rearranged. This gives the identity.  $\square$

At the Liouville fibre and for  $\Re s > 1$ ,

$$\sum_{n \geq 1} \frac{\lambda(n)}{n^s} = \prod_p (1 + p^{-s})^{-1} = \prod_p \frac{1 - p^{-s}}{1 - p^{-2s}} = \frac{\zeta(2s)}{\zeta(s)}.$$

### 13.4 Every completely multiplicative function is a fibre

**Theorem 13.8** (Universality of View A). *Let  $a : \mathbb{N} \rightarrow \mathbb{C}$  be completely multiplicative. For every prime  $p$ , choose a root  $\beta_p$  of*

$$f_p(X) - a(p).$$

*Then*

$$\boxed{F_n((\beta_p)_p) = a(n)}$$



for every  $n \in \mathbb{N}$ .

**Finite computational check V28: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The polynomial  $f_p(X) - a(p)$  is nonconstant and therefore has a complex root. By complete multiplicativity,

$$F_n((\beta_p)_p) = \prod_p f_p(\beta_p)^{v_p(n)} = \prod_p a(p)^{v_p(n)} = a(n).$$

□

Thus the root-evaluation family contains, as fibres, the Liouville function, Dirichlet characters, twists  $n^{it}$ , and many other completely multiplicative functions. Different choices of roots may correspond to different points of the parameter space with the same arithmetic fibre.

## 14 Application IV: Galois groups and monodromy

The polynomial family also supplies a computable branched-cover problem. For a prime  $p$ , one studies

$$f_p(X) - T$$

over  $\mathbb{Q}(T)$ . Three concrete outputs are available: the factorisation of  $p - 1$  forces a branch cycle over  $T = 1$ ; the discriminant detects critical-value collisions; and a functional decomposition proves that the monodromy action is imprimitive.

**Example 14.1** (A complete calculation for  $p = 19$ ). Since  $19 - 1 = 2 \cdot 3^2$ ,

$$f_{19}(X) = 1 + X(1 + X)^2 = X^3 + 2X^2 + X + 1.$$

Its derivative is

$$f'_{19}(X) = 3X^2 + 4X + 1 = (3X + 1)(X + 1).$$

The critical points  $-1$  and  $-1/3$  have distinct critical values

$$f_{19}(-1) = 1, \quad f_{19}(-1/3) = \frac{23}{27}.$$

Therefore  $f_{19}$  is Morse. Equivalently, up to a nonzero factor,

$$\Delta_{f_{19}}(T) = (T - 1)(27T - 23)$$

is squarefree. The Morse criterion below then proves that the geometric and arithmetic generic monodromy groups are both  $S_3$ .

For contrast,  $17 - 1 = 2^4$  gives  $f_{17}(X) = X^4 + 1$ . The fibre over  $T = 1$  is  $X^4 = 0$ , so the branch cycle over 1 is a 4-cycle and the discriminant has order 3 there. The polynomial is also decomposable through  $X^2$ . These two examples show how visible arithmetic structure leads to explicit monodromy information.

### 14.1 Generic monodromy and the discriminant test

The special polynomial  $f_p(X)$  corresponds to one value of a more informative one-parameter family. Let  $f \in \mathbb{Q}[X]$  have degree  $d \geq 2$ , and let  $T$  be transcendental.

**Definition 14.2** (Arithmetic and geometric monodromy). The *arithmetic monodromy group* of  $f$  is

$$G_f^{\text{ar}} := \text{Gal}(f(X) - T / \mathbb{Q}(T)),$$

acting on the  $d$  roots of  $f(X) - T$  in a splitting field. The *geometric monodromy group* is

$$G_f^{\text{geom}} := \text{Gal}(f(X) - T / \overline{\mathbb{Q}}(T)).$$

It is a normal subgroup of  $G_f^{\text{ar}}$ .

The polynomial  $f(X) - T$  is irreducible over  $\overline{\mathbb{Q}}(T)$ : viewed in  $\overline{\mathbb{Q}}[X, T]$ , it has degree one in  $T$ , so a nontrivial factorisation would force one factor to divide its constant leading coefficient in  $T$ . Hence both monodromy groups act transitively. A loop around  $T = \infty$  permutes the inverse branches as a  $d$ -cycle, because asymptotically the roots are proportional to the  $d$  choices of  $T^{1/d}$ .

The finite branch values are the critical values  $f(c)$ , where  $f'(c) = 0$ . This can be encoded in one polynomial.

**Definition 14.3** (Discriminant polynomial). Set

$$\Delta_f(T) := \text{Disc}_X(f(X) - T) \in \mathbb{Q}[T].$$

**Proposition 14.4** (Critical-value factorisation). *Up to a nonzero constant,*

$$\Delta_f(T) = \prod_{c: f'(c)=0} (T - f(c))^{\text{mult}_c(f')}.$$

*The product is over the distinct critical points of  $f$ .*

**Finite computational check V29: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* For a monic polynomial, the discriminant is, up to a sign, the resultant

$$\text{Res}_X(f(X) - T, f'(X)).$$

The product formula for the resultant evaluates the first polynomial at every root  $c$  of  $f'$ , with the multiplicity of that root. All leading-coefficient factors are nonzero constants independent of  $T$ .  $\square$

**Definition 14.5** (Morse polynomial). A polynomial  $f$  of degree  $d$  is called *Morse* if  $f'$  has  $d - 1$  distinct roots and the corresponding critical values are pairwise distinct.

Thus  $f$  is Morse exactly when  $\Delta_f(T)$  is squarefree.

**Theorem 14.6** (Morse criterion for full symmetric monodromy). *If  $f \in \mathbb{Q}[X]$  is Morse of degree  $d$ , then*

$$\boxed{G_f^{\text{geom}} = G_f^{\text{ar}} = S_d.}$$

**Finite computational check V30: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* At a simple critical point with a critical value not shared by another critical point, the local map has the form

$$T - f(c) = a(X - c)^2 + \text{higher terms}, \quad a \neq 0.$$

A small loop around that branch value therefore exchanges exactly two inverse branches and fixes the others; its local monodromy is a transposition. The geometric monodromy is transitive and generated by the local branch permutations. Form a graph whose  $d$  vertices are the sheets and whose edges are the transpositions occurring at finite branch values. The group generated by edge transpositions is the direct product of the symmetric groups on the connected components of this graph. Transitivity forces the graph to be connected, so these transpositions generate  $S_d$ . Hence  $G_f^{\text{geom}} = S_d$ , and the arithmetic group, being a subgroup of  $S_d$  containing the geometric group, is also  $S_d$ .  $\square$

This is a sufficient criterion, not a necessary one. A polynomial may fail to be Morse and still have full symmetric monodromy.

## 14.2 Translation, decomposition, and imprimitivity for Pratt polynomials

For every odd prime  $p$ ,

$$f_p(X) = f_{p-1}(X) + 1.$$

Consequently the covers defined by  $f_p$  and  $f_{p-1}$  differ only by translation of the target coordinate.

**Proposition 14.7** (Target translation). *For every odd prime  $p$ ,*

$$\boxed{\Delta_{f_p}(T) = \Delta_{f_{p-1}}(T - 1),}$$

*and  $f_p$  and  $f_{p-1}$  have isomorphic arithmetic and geometric monodromy groups.*

**Finite computational check V31: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* One has

$$f_p(X) - T = f_{p-1}(X) - (T - 1).$$

Replacing the transcendental target coordinate  $T$  by  $T - 1$  is a field automorphism and preserves the covering and its monodromy. The discriminant identity follows from the same substitution.  $\square$

**Proposition 14.8** (Decomposition is unchanged by the Pratt shift). *The polynomial  $f_p$  is functionally decomposable over a field  $K$  if and only if  $f_{p-1}$  is functionally decomposable over  $K$ . More precisely,*

$$f_{p-1} = g \circ h \iff f_p = (g + 1) \circ h.$$

**Finite computational check V32: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Adding or subtracting 1 from the outer polynomial gives the two implications.  $\square$

By the classical correspondence of Ritt, a nontrivial functional decomposition of a complex polynomial produces an imprimitive geometric monodromy action, and conversely an imprimitive block system produces an intermediate polynomial cover. Thus decomposition tests are a direct way to locate structured nonsymmetric monodromy.

For example,

$$\begin{aligned} f_{17}(X) &= X^4 + 1 = (Y^2 + 1) \circ X^2, \\ f_{37}(X) &= (X^2 + X)^2 + 1 = (Y^2 + 1) \circ (X^2 + X), \end{aligned}$$

and

$$f_{43}(X) = Y^2 + Y + 1 \mid_{Y=X^2+X}.$$

Their generic monodromy actions are therefore imprimitive.

### 14.3 A branch cycle forced by the factorisation of $p - 1$

The recursion gives more than a general research direction: it determines one finite branch cycle exactly.

Write

$$p - 1 = \prod_{q \mid p-1} q^{e_q}, \quad e_q = v_q(p - 1),$$

and put

$$d_q := \deg f_q = m_2(q) = C_{2,q}.$$

We use the established irreducibility of the prime-index polynomials  $f_q$ ; in characteristic zero they are separable, and distinct prime indices give distinct monic irreducible polynomials because  $f_q(2) = q$ .

**Theorem 14.9** (Ramification over the value 1). *For an odd prime  $p$ , local monodromy of the cover  $f_p : \mathbb{P}^1 \rightarrow \mathbb{P}^1$  around the value  $T = 1$  has cycle type*

$$\prod_{q: e_q \geq 2} (e_q)^{d_q} 1^{\sum_{q: e_q=1} d_q}.$$

*Equivalently,*

$$\text{ord}_{T=1} \Delta_{f_p}(T) = \sum_{q \mid p-1} (e_q - 1) d_q = \sum_{q \mid p-1} (v_q(p - 1) - 1) C_{2,q}.$$

**Finite computational check V33: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Set

$$g(X) = f_{p-1}(X) = \prod_{q|p-1} f_q(X)^{e_q}, \quad f_p(X) = 1 + g(X).$$

Let  $\alpha$  be a root of  $f_q$ . Pairwise coprimality and separability show that  $\alpha$  is a zero of  $g$  of multiplicity exactly  $e_q$ . Hence locally

$$f_p(X) - 1 = g(X) = a(X - \alpha)^{e_q} + \text{higher terms}, \quad a \neq 0.$$

The ramification index above 1 is therefore  $e_q$ . The polynomial  $f_q$  has  $d_q$  distinct roots, so it contributes  $d_q$  disjoint cycles of length  $e_q$  when  $e_q \geq 2$ , and  $d_q$  fixed sheets when  $e_q = 1$ . A ramification point of index  $e_q$  contributes  $e_q - 1$  to the discriminant order. Summing over all roots gives the formula.  $\square$

**Corollary 14.10** (A necessary condition for the Morse property). *If  $f_p$  is Morse, then all exponents  $e_q = v_q(p-1)$  equal 1, with at most one exception; the exceptional exponent may only be 2, and then  $m_2(q) = 1$ , hence  $q \in \{2, 3\}$ .*

**Finite computational check V34: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* For a Morse polynomial the discriminant is squarefree, so its order at  $T = 1$  is at most 1. The preceding theorem gives

$$\sum_q (e_q - 1)m_2(q) \leq 1.$$

Every summand is a nonnegative integer. Therefore at most one is nonzero, and that nonzero term must equal 1. Thus  $e_q = 2$  and  $m_2(q) = 1$ . The bound  $\log_3 q \leq m_2(q)$  then implies  $q \leq 3$ .  $\square$

The condition is not sufficient. Even when  $p - 1$  is squarefree, other critical points can collide in value. For instance,

$$f_{43}(X) = (X^2 + X)^2 + (X^2 + X) + 1$$

is decomposable and has a repeated critical value away from 1.

#### 14.4 Examples of discriminants and decompositions

The following table is obtained by direct differentiation and resultant calculation. It illustrates four different phenomena: Morse symmetric monodromy, repeated ramification at 1, functional decomposition, and non-Morse indecomposable behaviour.

| $p$ | $f_p(X)$                   | $\Delta_{f_p}(T)$ up to sign        | Morse? | decomposition               |
|-----|----------------------------|-------------------------------------|--------|-----------------------------|
| 5   | $X^2 + 1$                  | $4(T - 1)$                          | yes    | none                        |
| 19  | $X(X + 1)^2 + 1$           | $(T - 1)(27T - 23)$                 | yes    | none                        |
| 17  | $X^4 + 1$                  | $256(T - 1)^3$                      | no     | $(Y^2 + 1) \circ X^2$       |
| 37  | $(X^2 + X)^2 + 1$          | $16(T - 1)^2(16T - 17)$             | no     | $(Y^2 + 1) \circ (X^2 + X)$ |
| 43  | $(Y^2 + Y + 1) _{Y=X^2+X}$ | $(4T - 3)^2(16T - 13)$              | no     | quadratic after quadratic   |
| 41  | $X^5 + X^3 + 1$            | $(T - 1)^2(3125T^2 - 6250T + 3233)$ | no     | none                        |

The last row is important: failure of the Morse test does not imply decomposition or nonsymmetric monodromy. It only means that the simplest transposition criterion is unavailable.

## 14.5 View A as a product of branched covers

Let  $S$  be a finite set of primes. The coordinate map underlying View A is

$$\Pi_S : \prod_{p \in S} \mathbb{A}_{x_p}^1 \longrightarrow \prod_{p \in S} \mathbb{A}_{t_p}^1, \quad t_p = f_p(x_p).$$

Remove from each target line the branch values of  $f_p$ . Over the remaining product,  $\Pi_S$  is an unramified finite cover.

**Proposition 14.11** (Product monodromy). *The geometric monodromy group of  $\Pi_S$  is the direct product*

$$\prod_{p \in S} G_{f_p}^{\text{geom}}.$$

**Finite computational check V35: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* A loop moving only the coordinate  $t_p$  acts only on the inverse branches of  $f_p$  and fixes every other coordinate. Such coordinate loops generate the fundamental group of the product of punctured lines. Hence the monodromy actions commute and independently generate the direct product.  $\square$

The Liouville construction chooses one point in the special product fibre

$$t_p = -1 \quad (p \in S).$$

Generic monodromy controls how all possible root choices are permuted as the target values move, while the arithmetic Galois group of the fixed fibre  $f_p(X) + 1$  may be smaller. Hilbert irreducibility implies that many rational target values inherit the generic arithmetic group, but it does not force the particular values 0 or  $-1$  to be generic.

View B has a different algebraic character. The variables  $z_q$  are simultaneous label weights in a multivariate polynomial  $\Phi_n(\mathbf{z})$  rather than independent inverse coordinates of univariate covers. Natural monodromy problems for View B arise after choosing an algebraic slice

$$z_q = a_q X + b_q$$

and studying

$$\Phi_n((a_q X + b_q)_q) - T,$$

or by studying the discriminant variety of the multivariate map itself. The root-label master polynomial allows the same operation separately in each root block.

## 14.6 A concrete Galois–monodromy research programme

The preceding results suggest a sequence of tests that separates provable structure from numerical evidence.

1. Compute

$$\Delta_p(T) := \text{Disc}_X(f_p(X) - T).$$

Its factor at  $T = 1$  is predicted exactly by the theorem above.

2. Test whether  $\Delta_p$  is squarefree. A positive result proves

$$G_{f_p}^{\text{geom}} = S_{m_2(p)}.$$

3. If the discriminant is not squarefree, factor it to identify collisions of critical values and calculate the corresponding local branch-cycle types.
4. Test functional decomposition of  $f_p$ . A decomposition proves imprimitivity. The identity

$$f_p(X^2) = f_{f_p(4)}(X),$$

when it holds and the right index is prime, gives an explicit quadratic right component and hence an imprimitive tower, as observed in the MathOverflow discussion.

5. For the special fibre  $T = 0$ , determine  $\text{Gal}(f_p/\mathbb{Q})$  by irreducibility, discriminants, and factorisation patterns modulo good primes. Compare it with the generic group and with the Liouville fibre  $T = -1$ .
6. Compare exceptional branch cycles and decomposition degrees with the arithmetic data

$$(v_q(p-1))_q, \quad (C_{q,p})_q, \quad \sum_q (v_q(p-1) - 1)C_{2,q}.$$

This asks whether nonsymmetric groups are predicted by visible Pratt substructures.

The strongest asymptotic question remains open: whether the proportion of primes for which the special group  $\text{Gal}(f_p/\mathbb{Q})$  is the full symmetric group tends to 1. A more accessible intermediate problem is to estimate separately the primes for which  $f_p$  is Morse, the primes for which it is decomposable, and the primes for which it is indecomposable but has colliding critical values.

## 15 Tree inequalities and their Pratt-forest consequences

This section begins with inequalities valid for arbitrary finite rooted forests and then translates them into arithmetic statements for Pratt forests. No probabilistic or physical interpretation is needed.

### 15.1 Basic parameters of an abstract rooted forest

Let  $F$  be a finite rooted forest. Write

$$R(F) := \#\{\text{connected components of } F\},$$

$$L(F) := \#\{\text{leaves of } F\}, \quad I(F) := \#\{\text{non-leaf vertices of } F\},$$

and

$$V(F) := L(F) + I(F) = \#V(F).$$

For a vertex  $v$ , let  $d^+(v)$  denote its number of children. Finally, put

$$A(F) := \#CF(F).$$

For the empty forest we use  $R = L = I = V = 0$  and  $A = 1$ .

**Lemma 15.1** (Edge count). *A finite rooted forest satisfies*

$$\#E(F) = V(F) - R(F).$$

**Finite computational check V36: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Each connected component is a finite tree. A tree with  $v$  vertices has  $v - 1$  edges. Summing this identity over the  $R(F)$  components gives

$$\#E(F) = \sum_j (V_j - 1) = V(F) - R(F).$$

□

**Theorem 15.2** (Leaf-excess identity). *For every finite rooted forest,*

$$L(F) = R(F) + \sum_{v \text{ non-leaf}} (d^+(v) - 1).$$

**Finite computational check V37: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Every edge begins at exactly one non-leaf vertex, hence

$$\sum_{v \text{ non-leaf}} d^+(v) = \#E(F) = V(F) - R(F).$$

Subtracting  $I(F)$  and using  $V(F) = L(F) + I(F)$  gives

$$\sum_{v \text{ non-leaf}} (d^+(v) - 1) = V(F) - R(F) - I(F) = L(F) - R(F).$$

Rearranging proves the identity.

□



**Corollary 15.3.** *For every finite rooted forest,*

$$L(F) \geq R(F).$$

*If every non-leaf vertex has at least  $d \geq 2$  children, then*

$$L(F) \geq R(F) + (d - 1)I(F).$$

**Finite computational check V38: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Both claims follow directly from the leaf-excess identity because each summand  $d^+(v) - 1$  is nonnegative, and is at least  $d - 1$  under the additional hypothesis.  $\square$

## 15.2 Sizes and numbers of complete fronts

**Theorem 15.4** (Size of a complete front). *For every complete front  $A \in \text{CF}(F)$ ,*

$$R(F) \leq |A| \leq L(F).$$

**Finite computational check V39: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Every component contains a root-to-leaf path, and a complete front meets every such path. Hence  $A$  contains at least one vertex from each component, proving  $|A| \geq R(F)$ .

For the upper bound, choose for every  $a \in A$  one leaf descending from  $a$ . If  $a, b \in A$  are distinct, then they are incomparable. Their descendant subtrees are therefore disjoint, so the chosen descendant leaves are distinct. This gives an injection from  $A$  into the leaf set of  $F$ , and hence  $|A| \leq L(F)$ .  $\square$

**Theorem 15.5** (Complete-front count). *For every finite rooted forest,*

$$I(F) + 1 \leq A(F) \leq 2^{I(F)}.$$

**Finite computational check V40: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* We first treat one rooted tree  $T$  by induction on  $I(T)$ . If  $T$  is a single leaf, then  $I(T) = 0$  and  $A(T) = 1$ , so both bounds are equalities.

Suppose that the root is non-leaf and that its child subtrees are  $T_1, \dots, T_k$ . A complete front either consists of the root alone or avoids the root and chooses one complete front in every child subtree. Therefore

$$A(T) = 1 + \prod_{j=1}^k A(T_j), \quad I(T) = 1 + \sum_{j=1}^k I(T_j).$$

By induction,

$$A(T) \geq 1 + \prod_j (I(T_j) + 1) \geq 1 + \left(1 + \sum_j I(T_j)\right) = I(T) + 1.$$

Here we used  $\prod_j (1 + a_j) \geq 1 + \sum_j a_j$  for nonnegative  $a_j$ . Similarly,

$$A(T) \leq 1 + 2^{\sum_j I(T_j)} \leq 2^{1 + \sum_j I(T_j)} = 2^{I(T)}.$$

Now let  $F = T_1 \sqcup \dots \sqcup T_r$ . Complete-front choices in distinct components are independent, so

$$A(F) = \prod_{j=1}^r A(T_j), \quad I(F) = \sum_{j=1}^r I(T_j).$$

Consequently

$$A(F) \geq \prod_j (I(T_j) + 1) \geq 1 + \sum_j I(T_j) = I(F) + 1$$

and

$$A(F) \leq \prod_j 2^{I(T_j)} = 2^{I(F)}.$$

The empty forest also satisfies the statement. □

### 15.3 Conserved flows across complete fronts

The preceding bounds use only the cardinalities of fronts. A stronger principle is available: every complete front is a cut separating the roots from the leaves, and any additive flow through the tree is conserved across such a cut.

**Definition 15.6** (Front-conserved weight). Let  $F$  be a finite rooted forest. A function

$$w : V(F) \longrightarrow \mathbb{R}$$

is called *front-conserved* if every non-leaf vertex satisfies

$$w(v) = \sum_{u \text{ child of } v} w(u).$$

**Theorem 15.7** (Cut-conservation identity). *If  $w$  is front-conserved and  $A \in \text{CF}(F)$ , then*

$$\boxed{\sum_{v \in A} w(v) = \sum_{r \text{ root of } F} w(r).}$$

**Finite computational check V41: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* It suffices to prove the statement for one rooted tree. Begin with the root front  $\{r\}$ , whose total weight is  $w(r)$ . Replacing a selected non-leaf vertex  $v$  by all of its children preserves the total weight by the defining identity for  $w$ . Every complete front can be obtained from the root front by finitely many such refinements: if the root is absent, refine

it, and continue recursively only in the child subtrees whose roots are absent. Therefore every complete front has total weight  $w(r)$ . Summing over components proves the forest statement.  $\square$

**Corollary 15.8** (Descendant-leaf conservation). *For a vertex  $v$ , let  $\ell_F(v)$  be the number of leaves descending from  $v$ , including  $v$  itself when  $v$  is a leaf. Then every complete front  $A$  satisfies*

$$\sum_{v \in A} \ell_F(v) = L(F).$$

**Finite computational check V42: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* For a leaf,  $\ell_F(v) = 1$ . For a non-leaf vertex, the descendant leaf sets of its child subtrees are disjoint and exhaust its descendant leaves, so

$$\ell_F(v) = \sum_{u \text{ child of } v} \ell_F(u).$$

Thus  $\ell_F$  is front-conserved. Its sum over the roots is the total number of leaves of the forest.  $\square$

**Corollary 15.9** (A Kraft-type equality). *For each non-leaf vertex  $v$ , assign positive transition probabilities  $\pi(v, u)$  to its children, with*

$$\sum_{u \text{ child of } v} \pi(v, u) = 1.$$

*Give every root mass 1, and define the mass of a descendant recursively by*

$$\mu(u) = \mu(v)\pi(v, u).$$

*Then every complete front satisfies*

$$\sum_{v \in A} \mu(v) = R(F).$$

*In particular, choosing uniform transitions gives*

$$\mu(v) = \prod_{a \prec v} \frac{1}{d^+(a)},$$

*where the product runs over the proper ancestors of  $v$  in its component.*

**Finite computational check V43: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The mass function is front-conserved because the masses of the children of  $v$  sum to  $\mu(v)$ . Apply the cut-conservation theorem.  $\square$

**Proposition 15.10** (A componentwise lower bound for the number of fronts). *Let  $R_+(F)$  be the number of components that contain at least one non-leaf vertex. Then*

$$\boxed{A(F) \geq 2^{R_+(F)}}.$$

**Finite computational check V44: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* A nontrivial rooted tree has at least two distinct complete fronts: the singleton root and the set of all leaves. A one-vertex component has exactly one. Front choices in different components multiply, which proves the bound.  $\square$

**Proposition 15.11** (Uniform random-front bounds). *Choose a complete front uniformly from  $\text{CF}(F)$  and let  $Y = |A|$ . Then*

$$R(F) \leq \mathbb{E}[Y] \leq L(F), \quad \text{Var}(Y) \leq \frac{(L(F) - R(F))^2}{4}.$$

**Finite computational check V45: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The expectation bound follows from  $R(F) \leq Y \leq L(F)$ . For the variance, any random variable  $Y \in [a, b]$  satisfies

$$(Y - a)(b - Y) \geq 0.$$

Taking expectations and rearranging gives

$$\text{Var}(Y) \leq (\mathbb{E}[Y] - a)(b - \mathbb{E}[Y]) \leq \frac{(b - a)^2}{4}.$$

Apply this with  $a = R(F)$  and  $b = L(F)$ .  $\square$

## 15.4 Translation to Pratt forests

For  $n \geq 1$ , introduce the abbreviations

$$R_n := R(\mathcal{F}_n), \quad L_n := L(\mathcal{F}_n), \quad I_n := I(\mathcal{F}_n), \quad V_n := V(\mathcal{F}_n), \quad A_n := A(\mathcal{F}_n).$$

The definitions of a Pratt forest immediately give

$$R_n = \Omega(n), \quad L_n = m_2(n), \quad I_n = \sum_{q \geq 3} m_q(n), \quad V_n = \sum_q m_q(n),$$

and the complete-front theorem gives

$$A_n = f_n(1).$$

**Theorem 15.12** (Exact Pratt leaf identity). *For every  $n \geq 1$ ,*

$$m_2(n) = \Omega(n) + \sum_{q \geq 3} m_q(n)(\Omega(q-1) - 1).$$

**Finite computational check V46: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* A vertex labelled  $q > 2$  has exactly  $\Omega(q-1)$  children, counted with multiplicity, because its children are the prime factors of  $q-1$ . The vertices labelled 2 are exactly the leaves. Substitute these outdegrees into the abstract leaf-excess identity.  $\square$

**Corollary 15.13** (Leaf lower bounds). *For every  $n \geq 1$ ,*

$$m_2(n) \geq \Omega(n).$$

Moreover,

$$m_2(n) \geq \Omega(n) + \sum_{q \geq 5} m_q(n).$$

**Finite computational check V47: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Every summand in the exact identity is nonnegative. For  $q = 3$ , one has  $\Omega(q-1) - 1 = 0$ . If  $q \geq 5$  is prime, then  $q-1$  is an even integer at least 4, so  $\Omega(q-1) \geq 2$  and therefore  $\Omega(q-1) - 1 \geq 1$ .  $\square$

**Corollary 15.14** (Size of a complete Pratt front). *Every  $A \in \text{CF}(\mathcal{F}_n)$  satisfies*

$$\Omega(n) \leq |A| \leq m_2(n).$$

*Equivalently, every exponent vector occurring in the master polynomial satisfies*

$$\Omega(n) \leq \sum_{p,q} N_{p,q}(A) \leq m_2(n).$$

**Finite computational check V48: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

**Corollary 15.15** (Number of complete Pratt fronts). *For every  $n \geq 1$ ,*

$$1 + \sum_{q \geq 3} m_q(n) \leq f_n(1) \leq 2 \sum_{q \geq 3} m_q(n).$$

**Finite computational check V49: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Apply the abstract complete-front count to  $\mathcal{F}_n$  and use  $I_n = \sum_{q \geq 3} m_q(n)$  and  $A_n = f_n(1)$ .  $\square$

**Theorem 15.16** (Weighted conservation for Pratt fronts). *For every complete front  $A \in \text{CF}(\mathcal{F}_n)$ ,*

$$\sum_{p,q} m_2(q) N_{p,q}(A) = m_2(n).$$

*More precisely, for each root type  $p$ ,*

$$\sum_q m_2(q) N_{p,q}(A) = v_p(n) m_2(p).$$

**Finite computational check V50: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* A vertex labelled  $q$  is the root of a descendant subtree isomorphic to  $T_q$ , and that subtree has exactly  $m_2(q)$  leaves. Apply descendant-leaf conservation separately to the union of the  $v_p(n)$  components rooted at  $p$ . Summing over  $p$  gives the first identity.  $\square$

**Corollary 15.17** (Weighted homogeneity). *Assign weight*

$$\text{wt}(z_q) = m_2(q).$$

*Then  $\Phi_n(\mathbf{z})$  is weighted-homogeneous of weighted degree  $m_2(n)$ . Moreover, the master polynomial is separately weighted-homogeneous in every root block:*

$$\text{wt}_p(z_{r,q}) = \begin{cases} m_2(q), & r = p, \\ 0, & r \neq p, \end{cases}$$

*and its  $p$ -block weighted degree is  $v_p(n) m_2(p)$ .*

**Finite computational check V51: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The exponent vector of the monomial associated with  $A$  is  $(N_q(A))_q$  for  $\Phi_n$  and  $(N_{p,q}(A))_{p,q}$  for  $\mathcal{M}_n$ . The preceding theorem says that every monomial has the asserted weighted degree.  $\square$

**Corollary 15.18** (An additional front-count lower bound). *For every  $n \geq 1$ ,*

$$f_n(1) \geq 2^{\Omega(n) - v_2(n)}.$$

**Finite computational check V52: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The nontrivial components of  $\mathcal{F}_n$  are exactly those rooted at odd primes. Their number, counted with multiplicity, is  $\Omega(n) - v_2(n)$ . Apply the abstract componentwise bound.  $\square$

## 15.5 Bounds for the univariate front polynomial

**Proposition 15.19** (Range of the terminal count). *For every complete front  $A$  of  $\mathcal{F}_n$ ,*

$$v_2(n) \leq N_2(A) \leq m_2(n).$$

**Finite computational check V53: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The forest contains  $v_2(n)$  isolated components equal to  $T_2$ . Their unique vertices must be selected by every complete front, giving the lower bound. The upper bound is immediate because the whole forest contains exactly  $m_2(n)$  vertices labelled 2.  $\square$

**Theorem 15.20** (Polynomial comparison inequalities). *For  $x \geq 1$ ,*

$$x^{v_2(n)} f_n(1) \leq f_n(x) \leq x^{m_2(n)} f_n(1).$$

*For  $0 < x \leq 1$ , the inequalities reverse:*

$$x^{m_2(n)} f_n(1) \leq f_n(x) \leq x^{v_2(n)} f_n(1).$$

**Finite computational check V54: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Use

$$f_n(x) = \sum_{A \in \text{CF}(\mathcal{F}_n)} x^{N_2(A)}$$

and apply the preceding bounds to each exponent. When  $x \geq 1$ , the function  $j \mapsto x^j$  is increasing; when  $0 < x \leq 1$ , it is decreasing.  $\square$

**Corollary 15.21** (Bounds for the master polynomial). *Assume that every active variable  $z_{p,q}$  lies in an interval  $[a, b]$ .*

*If  $1 \leq a \leq b$ , then*

$$f_n(1)a^{\Omega(n)} \leq \mathcal{M}_n(\mathbf{z}) \leq f_n(1)b^{m_2(n)}.$$

*If  $0 < a \leq b \leq 1$ , then*

$$f_n(1)a^{m_2(n)} \leq \mathcal{M}_n(\mathbf{z}) \leq f_n(1)b^{\Omega(n)}.$$

**Finite computational check V55: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* A monomial belonging to a front  $A$  is a product of exactly  $|A|$  active variables. Use

$$\Omega(n) \leq |A| \leq m_2(n)$$

and sum the resulting monomial bounds over the  $f_n(1)$  complete fronts.  $\square$

**Corollary 15.22** (Uniform terminal-count moments). *For the uniform distribution on complete fronts, let  $X_n = N_2(A)$ . Then*

$$v_2(n) \leq \mathbb{E}[X_n] \leq m_2(n), \quad \text{Var}(X_n) \leq \frac{(m_2(n) - v_2(n))^2}{4}.$$

**Finite computational check V56: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Apply the same bounded-variable argument as in the abstract random-front proposition to

$$v_2(n) \leq X_n \leq m_2(n).$$

□

**Corollary 15.23** (Front-count bounds from evaluation at 2). *For every  $n \geq 1$ ,*

$$\boxed{\frac{n}{2^{m_2(n)}} \leq f_n(1) \leq \frac{n}{2^{v_2(n)}}.}$$

Consequently,

$$\boxed{\max \left\{ 1 + \sum_{q \geq 3} m_q(n), \frac{n}{2^{m_2(n)}} \right\} \leq f_n(1)}$$

and

$$\boxed{f_n(1) \leq \min \left\{ 2^{\sum_{q \geq 3} m_q(n)}, \frac{n}{2^{v_2(n)}} \right\}.}$$

**Finite computational check V57: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Set  $x = 2$  in the polynomial comparison inequalities and use  $f_n(2) = n$ . Combining the result with the abstract complete-front bound gives the final pair. □

**Corollary 15.24** (Degree and leading coefficient). *For every  $n \geq 1$ ,*

$$\deg f_n = m_2(n),$$

and  $f_n$  is monic.

**Finite computational check V58: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The set of all leaves of  $\mathcal{F}_n$  is a complete front and contributes the monomial  $x^{m_2(n)}$ . No complete front can select more than all leaves. Moreover, the leaf set is the unique complete front selecting every leaf, so the coefficient of the highest power is 1. □



## 15.6 Logarithmic, size, and height bounds

**Theorem 15.25** (Sharp logarithmic bounds for the number of leaves). *For every  $n \geq 1$ ,*

$$\log_3 n \leq m_2(n) \leq \log_2 n.$$

*The upper bound is attained by powers of 2, and the lower bound is attained by powers of 3.*

**Finite computational check V59: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The leaf front contributes the term  $2^{m_2(n)}$  to the positive sum  $f_n(2) = n$ . Therefore

$$2^{m_2(n)} \leq n,$$

which gives  $m_2(n) \leq \log_2 n$ .

For the other direction, we prove  $n \leq 3^{m_2(n)}$  by strong induction on  $n$ . It is true for  $n = 1$  and  $n = 2$ . If  $n = ab$  is composite with  $1 < a, b < n$ , then complete additivity gives

$$n = ab \leq 3^{m_2(a)} 3^{m_2(b)} = 3^{m_2(n)}.$$

Now let  $n = p$  be an odd prime. The 2-labelled vertices of  $T_p$  are exactly those in the descendant forest  $\mathcal{F}_{p-1}$ , so

$$m_2(p) = m_2(p-1).$$

By induction,

$$p-1 \leq 3^{m_2(p)}.$$

The left side is even and the right side is odd, hence in fact

$$p-1 \leq 3^{m_2(p)} - 1,$$

which implies  $p \leq 3^{m_2(p)}$ . This completes the induction.

Finally, complete additivity gives  $m_2(2^k) = k$  and  $m_2(3^k) = k$ , proving sharpness.  $\square$

**Corollary 15.26** (A multiplicative inequality for every complete front). *For  $A \in \text{CF}(\mathcal{F}_n)$  define the label product*

$$P(A) := \prod_{v \in A} \ell(v) = \prod_q q^{N_q(A)}.$$

*Then*

$$2^{m_2(n)} \leq P(A) \leq 3^{m_2(n)}.$$

*Both constants are sharp: the lower equality is attained by the leaf front, and the upper equality is attained by the root front of  $\mathcal{F}_{3^k}$ .*

**Finite computational check V60: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Weighted conservation gives

$$m_2(n) = \sum_q N_q(A) m_2(q).$$

The sharp bounds already proved for each prime label  $q$  are

$$\log_3 q \leq m_2(q) \leq \log_2 q.$$

Consequently,

$$\sum_q N_q(A) \log_3 q \leq m_2(n) \leq \sum_q N_q(A) \log_2 q.$$

Exponentiation gives the result. The leaf front has product  $2^{m_2(n)}$ . For  $n = 3^k$ , the root front has product  $3^k$  and  $m_2(3^k) = k$ .  $\square$

**Lemma 15.27** (Elementary product inequality). *If  $a_1, \dots, a_r$  are integers at least 2, then*

$$1 + \sum_{j=1}^r (a_j - 1) \leq \prod_{j=1}^r a_j.$$

**Finite computational check V61: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* For two factors,

$$ab - (1 + (a - 1) + (b - 1)) = (a - 1)(b - 1) \geq 0.$$

Induction on the number of factors proves the general statement.  $\square$

**Theorem 15.28** (Total number of vertices). *For every  $n \geq 2$ ,*

$$V_n = \sum_q m_q(n) \leq n - 1.$$

*In particular, for every prime  $p$ ,*

$$\sum_q C_{q,p} \leq p - 1.$$

**Finite computational check V62: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Let  $s(p) := V(T_p) = \sum_q m_q(p)$ . We prove  $s(p) \leq p - 1$  by induction over primes. For  $p = 2$ ,  $s(2) = 1$ . If  $p > 2$ , then

$$s(p) = 1 + \sum_{q|p-1} v_q(p-1)s(q).$$

By induction,

$$s(p) \leq 1 + \sum_{q|p-1} v_q(p-1)(q-1).$$

Apply the elementary product inequality to the prime factors of  $p - 1$ , repeated with

multiplicity. It gives

$$1 + \sum_{q|p-1} v_q(p-1)(q-1) \leq p-1.$$

Thus  $s(p) \leq p-1$ .

For arbitrary  $n \geq 2$ ,

$$V_n = \sum_p v_p(n)s(p) \leq \sum_p v_p(n)(p-1).$$

A second application of the same product inequality to the prime factors of  $n$ , repeated with multiplicity, yields

$$1 + \sum_p v_p(n)(p-1) \leq n.$$

Therefore  $V_n \leq n-1$ . □

**Definition 15.29** (Height). The height  $h(T)$  of a rooted tree is the maximum number of edges in a root-to-leaf path. The height of a nonempty rooted forest is the maximum height of its components.

**Theorem 15.30** (Height bound). *For every prime  $p$ ,*

$$h(T_p) \leq \log_2(p+1) - 1.$$

*If  $n \geq 2$  and  $P^+(n)$  denotes its largest prime factor, then*

$$h(\mathcal{F}_n) \leq \log_2(P^+(n)+1) - 1 \leq \log_2(n+1) - 1.$$

**Finite computational check V63: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* If  $p = 2$ , then  $h(T_2) = 0$  and the assertion is immediate. Assume  $p > 2$  and consider a longest label chain

$$p = p_0 > p_1 > \cdots > p_h = 2, \quad p_{j+1} \mid p_j - 1.$$

Then  $h \geq 1$ , and the final predecessor satisfies  $p_{h-1} \geq 3$ . At every earlier step,  $p_{j+1}$  is odd, while  $p_j - 1$  is even and divisible by  $p_{j+1}$ ; hence it is divisible by  $2p_{j+1}$  and

$$p_j \geq 2p_{j+1} + 1.$$

Iterating from  $p_{h-1} \geq 3$  gives

$$p \geq 2^{h+1} - 1.$$

Solving for  $h$  proves the prime bound. The height of a Pratt forest is the maximum of the heights of its root components, so the largest prime factor gives the first forest bound, and  $P^+(n) \leq n$  gives the second. □

## 15.7 Consequences for the incidence matrix $C$

For a prime  $p$ , the  $p$ th column of  $C$  is the node-count vector of  $T_p$ . Therefore

$$C_{2,p} = L(T_p), \quad \sum_{q \geq 3} C_{q,p} = I(T_p), \quad \sum_q C_{q,p} = V(T_p).$$

The preceding results immediately imply the following collection.

**Corollary 15.31** (Column inequalities for  $C$ ). *For every prime  $p$ ,*

$$\log_3 p \leq C_{2,p} \leq \log_2 p,$$

$$\sum_q C_{q,p} \leq p - 1,$$

and

$$1 + \sum_{q \geq 3} C_{q,p} \leq f_p(1) \leq 2^{\sum_{q \geq 3} C_{q,p}}.$$

Moreover,

$$C_{2,p} = 1 + \sum_{q \geq 3} C_{q,p} (\Omega(q - 1) - 1).$$

**Finite computational check V64: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Apply the logarithmic leaf bound, the total-vertex bound, the complete-front count, and the exact Pratt leaf identity to the single tree  $T_p$ . Since  $p$  has exactly one root component,  $\Omega(p) = 1$ .  $\square$

**Remark 15.32.** These inequalities separate three levels of information. The quantities  $C_{q,p}$  measure the deterministic capacity of the whole tree. The monomials of  $\Phi_p$  describe which label populations can occur simultaneously in one complete front. The value  $f_p(1)$  counts those fronts. Thus the abstract tree bounds constrain both the incidence matrix and the front-generating polynomials without identifying them.

## 16 Root fields, Pratt fields, and multivariate reconstruction

### 16.1 Linear transport of fields

Suppose a root-prime type  $p$  is assigned a scalar  $h_p$ . Define the root-level functional

$$E_{\text{root}}(n) := \sum_p h_p v_p(n).$$

The word “energy” may be used for  $E_{\text{root}}$ , but mathematically it is simply a linear functional on the ordinary valuation vector.

**Theorem 16.1** (Field transformation). *On any finite prime truncation, define*

$$\boldsymbol{\eta} := C^{-T} \mathbf{h} = (I - R^T) \mathbf{h}.$$

Then

$$\sum_p h_p v_p(n) = \sum_q \eta_q m_q(n).$$

Moreover,

$$\eta_q = h_q - \sum_{r|q-1} v_r(q-1)h_r.$$

**Finite computational check V65: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Since  $m = Cv$ , one has  $v = C^{-1}m$ . Hence

$$h^T v = h^T C^{-1}m = (C^{-T}h)^T m = \eta^T m.$$

The component formula follows from  $C^{-T} = I - R^T$  and  $R_{r,q} = v_r(q-1)$ .  $\square$

## 16.2 Multivariate Pratt reconstruction for View A

For the root-evaluation family set

$$h_p(\mathbf{x}) := \log f_p(x_p)$$

when a compatible logarithm is chosen. To avoid branch issues, the next theorem is stated directly as a rational-function identity.

**Theorem 16.2** (Multivariate Pratt reconstruction). *Define*

$$\Delta_2(\mathbf{x}) := f_2(x_2) = x_2,$$

and for an odd prime  $q$ ,

$$\Delta_q(\mathbf{x}) := \frac{f_q(x_q)}{\prod_{r|q-1} f_r(x_r)^{v_r(q-1)}}.$$

Then in the rational function field generated by the variables  $(x_p)_p$ ,

$$F_n(\mathbf{x}) = \prod_{q \in \mathbb{P}} \Delta_q(\mathbf{x})^{m_q(n)}.$$

**Finite computational check V66: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Take the exponent of a fixed factor  $f_r(x_r)$  on the right. It receives  $+m_r(n)$  from the numerator of  $\Delta_r$ . It receives

$$-\sum_{q>r} m_q(n)v_r(q-1)$$

from denominators. By  $v = C^{-1}m = (I - R)m$ , this total exponent equals  $v_r(n)$ . Hence the product equals

$$\prod_r f_r(x_r)^{v_r(n)} = F_n(\mathbf{x}).$$

□

**Corollary 16.3** (Diagonal local increments). *If  $x_p = x$  for every prime, then*

$$\Delta_q((x)_p) = \frac{f_q(x)}{f_{q-1}(x)}$$

and

$$f_n(x) = \prod_q \left( \frac{f_q(x)}{f_{q-1}(x)} \right)^{m_q(n)}.$$

For positive  $x$ , taking logarithms gives

$$\log f_n(x) = \sum_q m_q(n) \varepsilon_q(x), \quad \varepsilon_q(x) := \log \frac{f_q(x)}{f_{q-1}(x)}.$$

**Finite computational check V67: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Under diagonal substitution,

$$\prod_{r|q-1} f_r(x)^{v_r(q-1)} = f_{q-1}(x).$$

The other identities follow from the theorem. □

At  $x = 2$ ,

$$\varepsilon_q(2) = \log \frac{q}{q-1}, \quad \log n = \sum_q m_q(n) \log \frac{q}{q-1}.$$

This is an exact arithmetic identity, not a physical assumption.

## 17 Positive root-label fields and finite probability models

### 17.1 The parameter space

Fix  $n$ . Let

$$I_n := \{(p, q) \in \mathbb{P}^2 : v_p(n) > 0, C_{q,p} > 0\}.$$

This is finite.

**Definition 17.1** (Positive root-label parameter space). The positive parameter space is

$$\mathcal{Z}_n := (0, \infty)^{I_n}.$$

Its logarithmic coordinate space is

$$\Theta_n := \mathbb{R}^{I_n}, \quad z_{p,q} = e^{\theta_{p,q}}.$$

The coordinates  $\theta_{p,q}$  are called *root-label fields*.

The term “field” means only a real parameter coupled linearly to a counting statistic.

## 17.2 Partition function and Gibbs distribution

**Definition 17.2** (Log-partition function). Define

$$\psi_n(\boldsymbol{\theta}) := \log \mathcal{M}_n((e^{\theta_{p,q}})_{p,q}).$$

**Definition 17.3** (Exponential-family probability). For  $A \in \text{CF}(\mathcal{F}_n)$ , set

$$\mathbb{P}_{n,\boldsymbol{\theta}}(A) := \exp \left( \sum_{p,q} \theta_{p,q} N_{p,q}(A) - \psi_n(\boldsymbol{\theta}) \right).$$

**Proposition 17.4.** *The preceding formula defines a probability distribution on the finite set  $\text{CF}(\mathcal{F}_n)$ .*

**Finite computational check V68: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Every weight is positive. Summing over  $A$  gives

$$e^{-\psi_n(\boldsymbol{\theta})} \sum_A e^{\sum_{p,q} \theta_{p,q} N_{p,q}(A)} = e^{-\psi_n(\boldsymbol{\theta})} \mathcal{M}_n(e^\theta) = 1.$$

□

One may formally define an energy

$$E_{\boldsymbol{\theta}}(A) := - \sum_{p,q} \theta_{p,q} N_{p,q}(A).$$

Then  $\mathbb{P}(A) \propto e^{-E(A)}$ . This is exactly the algebraic form of a finite Gibbs ensemble, but it is used here only as a probability parametrisation.

## 17.3 Derivatives, covariance, and Fisher information

**Theorem 17.5** (First and second derivative identities). *For  $(p, q), (r, s) \in I_n$ ,*

$$\frac{\partial \psi_n}{\partial \theta_{p,q}} = \mathbb{E}_{n,\boldsymbol{\theta}}[N_{p,q}],$$

and

$$\frac{\partial^2 \psi_n}{\partial \theta_{p,q} \partial \theta_{r,s}} = \text{Cov}_{n,\boldsymbol{\theta}}(N_{p,q}, N_{r,s}).$$

**Finite computational check V69: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Since the sample space is finite, differentiation may be performed term by term. Let

$$Z(\theta) = \mathcal{M}_n(e^\theta) = \sum_A e^{\theta \cdot N(A)}.$$

Then

$$\partial_i \log Z = \frac{\sum_A N_i(A) e^{\theta \cdot N(A)}}{Z} = \mathbb{E}_\theta[N_i].$$

Differentiating once more,

$$\partial_j \mathbb{E}_\theta[N_i] = \mathbb{E}_\theta[N_i N_j] - \mathbb{E}_\theta[N_i] \mathbb{E}_\theta[N_j],$$

which is the covariance. □

**Corollary 17.6** (Positive semidefiniteness). *The Hessian matrix*

$$g_n(\theta) := \nabla^2 \psi_n(\theta)$$

*is positive semidefinite.*

**Finite computational check V70: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* For any real vector  $u = (u_i)$ ,

$$u^T g_n u = \text{Var}_\theta \left( \sum_i u_i N_i \right) \geq 0.$$

□

**Proposition 17.7** (Criterion for positive definiteness). *The matrix  $g_n(\theta)$  is positive definite if and only if no nonzero linear combination of the statistics  $N_{p,q}$  is constant on all complete fronts.*

**Finite computational check V71: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* The equality  $u^T g_n u = 0$  holds exactly when the random variable  $\sum_i u_i N_i$  has zero variance, which is equivalent to being constant on the support. Every front has positive probability, so the support is the whole set of complete fronts. □

**Theorem 17.8** (Forced null directions from tree conservation). *For every root type  $p$  with  $v_p(n) > 0$ , define a vector  $u^{(p)}$  on the active root-label coordinates by*

$$u_{r,q}^{(p)} := \begin{cases} m_2(q), & r = p, \\ 0, & r \neq p. \end{cases}$$

Then

$$\boxed{g_n(\theta) u^{(p)} = 0}$$



for every parameter value  $\theta$ . Consequently,

$$\text{rank } g_n(\theta) \leq |I_n| - \#\{p : v_p(n) > 0\}.$$

For View B, the vector  $(m_2(q))_q$  is always a null direction of the label covariance matrix.

**Finite computational check V72: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* Weighted front conservation gives

$$\sum_q m_2(q) N_{p,q}(A) = v_p(n) m_2(p),$$

which is constant on all complete fronts. Therefore its covariance with every statistic  $N_{r,s}$  is zero. This says exactly that the covariance Hessian annihilates  $u^{(p)}$ . The vectors belonging to distinct root blocks have disjoint supports and are linearly independent, proving the rank bound. Summing the block identities over  $p$  gives the View B statement.  $\square$

**Remark 17.9.** Thus the full root-label parameter space is never identifiable as written. The natural information-geometric manifold is obtained only after quotienting by these conserved directions or fixing one gauge condition in each root block.

If degeneracies occur, one obtains a genuine Riemannian metric by restricting to an identifiable linear subspace or by quotienting out the constant-statistic directions. The covariance matrix is also the Fisher information matrix of this finite exponential family.

## 17.4 View A and View B as pullback geometries

Let  $g^{\text{master}}$  denote the covariance metric on  $\Theta_n$ .

For View A, introduce coordinates  $\xi_p = \log x_p$  and embed

$$i_A : (\xi_p)_p \mapsto \theta_{p,q} = \begin{cases} \xi_p, & q = 2, \\ 0, & q > 2. \end{cases}$$

The pullback metric is the covariance matrix of the root-separated terminal counts  $N_{p,2}$ .

For View B, introduce label fields  $\eta_q$  and embed

$$i_B : (\eta_q)_q \mapsto \theta_{p,q} = \eta_q.$$

Since  $N_q = \sum_p N_{p,q}$ , the pullback metric is

$$(i_B^* g^{\text{master}})_{q,r} = \text{Cov}(N_q, N_r).$$

Thus the two geometries are not competing definitions; they are different lower-dimensional slices of one finite statistical model.

## 18 Canonical Gram matrices and modular language

### 18.1 Covariance as a Gram matrix

Let  $\tilde{N}_i = N_i - \mathbb{E}[N_i]$ . In the Hilbert space

$$L^2(\text{CF}(\mathcal{F}_n), \mathbb{P}_{n,\theta}),$$

one has

$$\text{Cov}(N_i, N_j) = \langle \tilde{N}_i, \tilde{N}_j \rangle_{L^2}.$$

Therefore the Fisher-covariance matrix is literally a Gram matrix of centred counting functions.

### 18.2 Gram matrices built directly from $C$

Let  $W = \text{diag}(w_q)$  with  $w_q > 0$  on a finite prime truncation. Define

$$G_{\text{root}} := C^T W C.$$

Then

$$G_{\text{root}}(p, r) = \sum_q w_q m_q(p) m_q(r).$$

It is the Gram matrix of the weighted Pratt-coordinate columns  $W^{1/2} C_{\bullet, p}$ .

Similarly, for a diagonal matrix  $D = \text{diag}(d_p)$  with  $d_p \geq 0$ , define

$$G_{\text{label}} := C D C^T.$$

Then

$$G_{\text{label}}(q, r) = \sum_p d_p m_q(p) m_r(p).$$

**Proposition 18.1.** *Both  $G_{\text{root}}$  and  $G_{\text{label}}$  are positive semidefinite. If  $W$  is positive definite and the finite truncation of  $C$  is square, then  $G_{\text{root}}$  is positive definite.*

**Finite computational check V73: PASS.** The tested range, method, package, and any numerical residual are recorded in Appendix A. This finite check supports error detection and reproducibility; it does not replace the proof or establish an asymptotic claim.

*Proof.* For every vector  $u$ ,

$$u^T C^T W C u = (C u)^T W (C u) \geq 0.$$

If  $W > 0$  and  $C$  is invertible, equality implies  $C u = 0$  and hence  $u = 0$ . Likewise,

$$u^T C D C^T u = (C^T u)^T D (C^T u) \geq 0.$$

□

### 18.3 Density matrices and modular operators

Given a nonzero positive semidefinite matrix  $G$ , define

$$\rho_G := \frac{G}{\text{Tr } G}.$$

Then  $\rho_G \geq 0$  and  $\text{Tr } \rho_G = 1$ . On the support of  $\rho_G$ , define

$$K_G := -\log \rho_G.$$

This is a self-adjoint matrix or operator. If a full-rank matrix is desired, replace  $G$  by  $G + \delta I$  with  $\delta > 0$  before normalisation.

The term *modular Hamiltonian* for  $K_G$  is standard mathematical language in quantum information and operator theory. In this article it does not imply that  $K_G$  governs a real time evolution. The entropy

$$S(\rho_G) := -\text{Tr}(\rho_G \log \rho_G)$$

is an information-theoretic invariant of the chosen Gram construction.

## 19 Further applications and computational recipes

The following applications use the same proved identities; they do not require additional physical assumptions.

### 19.1 Optimising front statistics by Newton polytopes

The exponent vectors of  $\Phi_n$  are the possible selected-label count vectors of complete fronts. Their convex hull is the Newton polytope  $\text{Newt}(\Phi_n)$ . Since forest union multiplies generating polynomials,

$$\text{Newt}(\Phi_n) = \sum_p v_p(n) \text{Newt}(\Phi_p),$$

a Minkowski sum.

Given real costs  $a_q$ , the largest possible value of

$$\sum_q a_q N_q(A)$$

over all complete fronts is the support function of this Newton polytope. It can be recovered from the tropical limit

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log \Phi_n((e^{ta_q})_q).$$

Thus questions such as “which complete front maximises a weighted label score?” become convex optimisation problems. The coordinate choices  $a_q = \delta_{q,r}$  recover the capacity  $m_r(n)$ .

### 19.2 Independent sums, central limit theorems, and rare fronts

The identity

$$\mathcal{M}_{ab} = \mathcal{M}_a \mathcal{M}_b$$

with independent component variables implies that additive front-count vectors for a product are sums of independent vectors. In particular,

$$\mathcal{F}_{n^k} = \underbrace{\mathcal{F}_n \sqcup \cdots \sqcup \mathcal{F}_n}_{k \text{ copies}}.$$

A uniformly random front of  $\mathcal{F}_{n^k}$  therefore produces a sum of  $k$  independent copies of the front-count vector for  $n$ . Whenever the covariance is nonzero on a chosen quotient space, the usual multivariate central limit theorem gives Gaussian fluctuations of order  $\sqrt{k}$ . The logarithmic moment-generating function also gives a Cramer-type rate function for exponentially rare front profiles.

### 19.3 A Pratt deformation of the zeta function

For fixed real  $x > 1$ , define

$$Z_x(s) := \sum_{n \geq 1} f_n(x)^{-s}.$$

Complete multiplicativity of  $f_n(x)$  gives

$$Z_x(s) = \prod_p (1 - f_p(x)^{-s})^{-1}$$

in the region of absolute convergence. At  $x = 2$ ,

$$f_n(2) = n, \quad Z_2(s) = \zeta(s).$$

This provides a precise multiplicative deformation of the Riemann zeta Dirichlet series. Concrete analytic tasks are to determine the counting function

$$N_x(Y) = \#\{n : f_n(x) \leq Y\},$$

the abscissa of convergence, and whether any theta-type transformation survives away from  $x = 2$ . No functional equation for  $x \neq 2$  is asserted here.

### 19.4 A reproducible computational pipeline

For a finite prime cutoff  $P$ , the following steps produce mutually checkable data.

1. Factor  $p - 1$  for every prime  $p \leq P$  and form the sparse immediate-child matrix  $R$ .
2. Compute the columns of  $C$  in increasing prime order using

$$C_{\bullet,p} = e_p + \sum_{q|p-1} v_q(p-1)C_{\bullet,q}.$$

3. Compute  $f_p$ ,  $\Phi_p$ , or selected slices of  $\mathcal{M}_p$  recursively. Verify

$$\deg_{z_q} \Phi_p = C_{q,p}.$$

4. For front statistics, evaluate derivatives at the neutral point. For exact distributions, extract coefficients.

5. For monodromy, compute  $f'_p$ , the discriminant  $\Delta_{f_p}(T)$ , and functional decompositions. Compare the factor at  $T = 1$  with

$$\sum_{q|p-1} (v_q(p-1) - 1)C_{2,q}.$$

6. For arithmetic fibres, choose target values  $a(p)$ , solve  $f_p(\beta_p) = a(p)$ , and evaluate partial sums of  $F_n((\beta_p))$ .

The degree identity, the matrix recursion, and direct tree counts give independent consistency checks. This makes the theory suitable for computer experiments in which errors in one representation can be detected by another.

### 19.5 Information geometry and positive matrices

For positive root-label variables, the normalised master polynomial defines a finite exponential family. The Hessian of its log-partition function is the covariance matrix and hence a Fisher information matrix. Separately,

$$C^T W C \quad \text{and} \quad C D C^T$$

are canonical positive semidefinite Gram matrices measuring, respectively, similarities between root trees and correlations between internal labels. After normalisation they may be studied using entropy, matrix logarithms, affine-invariant distances, and geodesics on positive-definite matrix spaces.

These constructions are useful as statistical and matrix-geometric invariants of recursive arithmetic data. They are not, without additional axioms and empirical content, models of spacetime, physical matter, or laboratory quantum systems.

### 19.6 Algebraic varieties of partial sums

The partial-sum polynomials

$$S_N(\mathbf{x}) = \sum_{n \leq N} F_n(\mathbf{x})$$

define hypersurfaces  $S_N(\mathbf{x}) = 0$ . Special points on these varieties encode cancellation of completely multiplicative functions. Natural calculations include factorisation, singular loci, intersections for several values of  $N$ , and the action of product monodromy on chosen fibres. These are concrete finite algebraic problems even though their asymptotic arithmetic interpretation may be difficult.

## 20 Open problems

**Question 20.1** (Geometry of the master Newton polytope). Can the face lattice of  $\text{Newt}(\mathcal{M}_n)$  be described directly from the ancestor poset of  $\mathcal{F}_n$ ?

**Question 20.2** (Exact rank of the Fisher metric). The weighted leaf-conservation laws provide one independent null direction for every active root block. Do these generate the entire kernel at generic positive field values? If not, which additional relations among the root-label counts  $N_{p,q}$  are forced by the branching structure?

**Question 20.3** (Recoverability). How much of the rooted Pratt forest can be reconstructed from  $\Phi_n$ , from  $F_n$ , or from the master polynomial  $\mathcal{M}_n$ ? The master polynomial contains strictly more labelled information than either marginal view, but complete injectivity has not been established here.

**Question 20.4** (Galois groups and monodromy). For how many primes  $p$  is the special group  $\text{Gal}(f_p(X)/\mathbb{Q})$  the full symmetric group? What is the density of primes for which  $f_p$  is Morse, and what is the density of primes for which  $f_p$  is functionally decomposable? Can every imprimitive family be explained by a recursive identity visible in the Pratt forest?

**Question 20.5** (Critical-value collisions). Beyond the forced branch value 1, can repeated factors of  $\Delta_{f_p}(T)$  be classified from identities among the descendant polynomials  $f_q$ ? In particular, can one predict the factorisation of the discriminant polynomial from the matrix  $C$  together with the exponent vector  $(v_q(p-1))_q$ ?

**Question 20.6** (Liouville fibre). How often does the special fibre  $f_p(X) + 1$  retain the generic monodromy group? Can the distribution of its roots, or the Galois action on the product fibre  $(f_p(X) + 1)_p$ , be related quantitatively to cancellation in the summatory Liouville function?

**Question 20.7** (Analytic continuation). For  $x \neq 2$ , what are the abscissa of convergence and analytic continuation properties of  $Z_x(s) = \sum_n f_n(x)^{-s}$ ? Is there any nontrivial duality replacing the Jacobi theta transformation at  $x = 2$ ?

**Question 20.8** (Limit geometry). Can the finite root-label parameter spaces be assembled into a controlled inductive limit as the prime cutoff grows? What weighted sequence spaces make the infinite matrices  $C$ ,  $C^T WC$ , or  $CDC^T$  bounded or trace class?

## 21 Conclusion

The usefulness of the root-label formalism is that one object answers several different questions.

From

$$\mathcal{M}_n((z_{p,q}))$$

one obtains exact complete-front distributions, expectations, covariances, conservation laws, and Newton polytopes. From

$$C_{q,p} = m_q(p)$$

one obtains a reusable coordinate algorithm

$$m(n) = C\mathbf{v}(n)$$

that avoids reconstructing every Pratt forest separately. From the root-evaluation family

$$F_n((x_p))$$

one obtains all completely multiplicative functions as algebraic fibres; the Liouville fibre converts the prime number theorem into a precise polynomial-fibre cancellation statement.

Finally, the recursion

$$f_p(X) - 1 = \prod_{q|p-1} f_q(X)^{v_q(p-1)}$$

turns the factorisation of  $p-1$  into explicit branching data for  $f_p(X)-T$ , while discriminants and decompositions provide practical monodromy tests.

The two marginal constructions therefore have complementary uses. View A is adapted to arithmetic fibres, Euler products, and Galois questions because it keeps root types separate. View B is adapted to front statistics because it keeps all selected labels. The variables  $z_{p,q}$  unite them without erasing either index.

The later probabilistic and matrix-geometric language is optional structure placed on top of these finite combinatorial facts. It supplies useful invariants, but the main mathematical contribution is already concrete: recursive trees are converted into calculable polynomials, coordinate transforms, probability laws, and branched covers.

## 22 A compact dictionary

| Object                     | Definition                                          | Meaning                                     |
|----------------------------|-----------------------------------------------------|---------------------------------------------|
| $v_p(n)$                   | Exponent of $p$ in $n$                              | Number of root components of type $p$       |
| $m_q(n)$                   | Number of $q$ -labelled vertices in $\mathcal{F}_n$ | Total recursive population of label $q$     |
| $C_{q,p}$                  | $m_q(p)$                                            | Number of $q$ -vertices inside $T_p$        |
| $R_{q,p}$                  | $v_q(p-1)$                                          | Immediate-child multiplicity                |
| $f_n(x)$                   | Recursive univariate polynomial                     | Generates selected terminal 2-counts        |
| $F_n((x_p))$               | $\prod_p f_p(x_p)^{v_p(n)}$                         | Root-separated evaluation, View A           |
| $\Phi_n((z_q))$            | $\sum_A \prod_q z_q^{N_q(A)}$                       | Fully label-marked front polynomial, View B |
| $\mathcal{M}_n((z_{p,q}))$ | $\sum_A \prod_{p,q} z_{p,q}^{N_{p,q}(A)}$           | Joint root-label master polynomial          |
| $\psi_n(\theta)$           | $\log \mathcal{M}_n(e^\theta)$                      | Log-partition function                      |
| $\nabla \psi_n$            | —                                                   | Mean root-label counts                      |
| $\nabla^2 \psi_n$          | —                                                   | Covariance/Fisher/Gram matrix               |

## A Computational verification ledger

This appendix records the finite test attached to each theorem-like statement. A **PASS** means that the supplied program completed the stated finite test without a counterexample and, for numerical tests, within the displayed residual. It does not mean that a universal or asymptotic theorem has been proved by computation.

The recorded software environment was Python 3 with SymPy 1.14.0, NetworkX 3.6.1, NumPy 2.3.5, SciPy 1.17.0, pandas 2.2.3, and mpmath 1.3.0. The random seed used for any

deterministic sample selection was 20260806; most checks are exhaustive in their stated range and use no randomness.

| ID  | Result                                                  | Finite scope                                        | Method and outcome                                                                                                     |
|-----|---------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| V01 | Complete additivity of Omega and multiplicity of lambda | all a,b $\leq 70$                                   | exact integer arithmetic with SymPy.factorint. Status: PASS.                                                           |
| V02 | Complete additivity of Pratt coordinates                | all a,b $\leq 70$                                   | exact recursive Pratt-coordinate dictionaries. Status: PASS.                                                           |
| V03 | Polynomial multiplicity and prime-shift recursion       | a,b $\leq 40$ ; primes $\leq 127$                   | exact symbolic polynomial identities in SymPy. Status: PASS.                                                           |
| V04 | Evaluation $f_n(2)=n$                                   | $1 \leq n \leq 500$                                 | exact symbolic evaluation. Status: PASS.                                                                               |
| V05 | Maximal antichains are complete                         | all unlabeled rooted forests with $\leq 7$ vertices | exhaustive NetworkX generation and subset enumeration. Status: PASS.                                                   |
| V06 | Extension of antichains to complete fronts              | same exhaustive forest family                       | every antichain tested for containment in a complete front. Status: PASS.                                              |
| V07 | Fully marked front recursion                            | primes $\leq 43$                                    | front enumeration compared with recursive symbolic formula. Status: PASS.                                              |
| V08 | Recovery of $f_n$ from the label-marked polynomial      | $1 \leq n \leq 60$                                  | exact specialisation of multivariate polynomials. Status: PASS.                                                        |
| V09 | Diagonal recovery of View A                             | $1 \leq n \leq 60$                                  | exact diagonal substitution. Status: PASS.                                                                             |
| V10 | Factorisation of the master polynomial                  | $1 \leq n \leq 40$                                  | master polynomial enumerated and compared with product factorisation. Status: PASS.                                    |
| V11 | Unification of Views A and B                            | $1 \leq n \leq 40$                                  | exact View A and View B substitutions. Status: PASS.                                                                   |
| V12 | Common one-dimensional slice                            | $1 \leq n \leq 40$                                  | exact common-slice substitution. Status: PASS.                                                                         |
| V13 | Coefficient and derivative extraction                   | $2 \leq n \leq 30$                                  | exact symbolic logarithmic derivatives versus enumerated moments. Status: PASS. Maximum numerical residual: 0.000e+00. |
| V14 | Ordinary-to-Pratt coordinate transform                  | $n \leq 500$ with largest prime factor $\leq 127$   | exact integer matrix multiplication. Status: PASS.                                                                     |
| V15 | Resolvent identity $C=(I-R)^{-1}$                       | prime truncation $\leq 127$                         | exact SymPy matrices and nilpotent geometric series. Status: PASS.                                                     |



| ID  | Result                                                   | Finite scope                                                    | Method and outcome                                                                                                            |
|-----|----------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| V16 | Weighted path formula for C                              | all $q, p \leq 127$                                             | matrix-power path counts. Status: PASS.                                                                                       |
| V17 | Equal labels form an antichain                           | Pratt trees $T_p$ for $p \leq 127$                              | explicit path comparison in recursively generated trees. Status: PASS.                                                        |
| V18 | Degree recovery of C from Phi                            | $p \leq 43$ and $n \leq 40$                                     | exact multivariate polynomial degrees. Status: PASS.                                                                          |
| V19 | Tropical recovery of C                                   | $p \leq 31$ , all active labels, $t=20$                         | numerical logarithmic derivative approaching the exact degree. Status: PASS. Maximum numerical residual: 8.245e-09.           |
| V20 | Recursive construction of columns of C                   | $p \leq 127$                                                    | exact recursive column construction. Status: PASS.                                                                            |
| V21 | Liouville specialisation                                 | 6 prime fibres; all supported $n \leq 200$                      | SymPy high-precision roots and numerical substitution. Status: PASS. Maximum numerical residual: 1.776e-15.                   |
| V22 | Landau equivalence (finite diagnostic only)              | partial sums through $N=20000$                                  | finite numerical diagnostic only; theorem remains analytic. Status: PASS. max ratios after 1000: M 0.01353, L 0.0371          |
| V23 | Convolution                                              | $1 \leq n \leq 500$                                             | exact divisor-sum convolution. Status: PASS.                                                                                  |
| V24 | Partial-sum convolution identities                       | $1 \leq x \leq 5000$                                            | exact integer partial-sum identities. Status: PASS.                                                                           |
| V25 | Equivalence of $M=o(x)$ and $L=o(x)$ (finite diagnostic) | $N \leq 20000$                                                  | parallel decay diagnostic for $M(N)/N$ and $L(N)/N$ ; not an asymptotic proof. Status: PASS. terminal ratios: 0.0013, -0.0027 |
| V26 | PNT as Liouville-fibre cancellation (finite diagnostic)  | $N \leq 20000$                                                  | Liouville fibre partial sums compared with $L(N)$ ; finite diagnostic only. Status: PASS. $L(N)/N=-0.0027$                    |
| V27 | Universal Euler product                                  | 4-prime Euler product, exponents 0..4                           | exact rational coefficient expansion of finite Euler factors. Status: PASS.                                                   |
| V28 | Universality of View A                                   | 4 arbitrary prime targets, exponents 0..3                       | numerical algebraic fibres from SymPy nroots. Status: PASS. Maximum numerical residual: 1.986e-15.                            |
| V29 | Critical-value factorisation of discriminants            | $f_p$ for $p$ in [5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43] | exact SymPy discriminant-resultant comparison. Status: PASS.                                                                  |

| ID  | Result                                                              | Finite scope                                                      | Method and outcome                                                                                                                                                                  |
|-----|---------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V30 | Morse criterion for symmetric monodromy (specialisation diagnostic) | Morse cases $[(7, 2, -7), (11, 3, -7), (19, 3, -7), (31, 4, -7)]$ | exact Morse tests plus SymPy number-field Galois groups of rational specialisations. Status: PASS. Finite Hilbert-specialisation evidence, not a computation over $\mathbb{Q}(T)$ . |
| V31 | Target translation of discriminants and monodromy data              | odd $p$ in $[5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43]$       | exact discriminant translation identity. Status: PASS.                                                                                                                              |
| V32 | Functional decomposition unchanged by Pratt shift                   | odd $p$ in $[5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43]$       | SymPy functional decomposition and exact shift reconstruction. Status: PASS.                                                                                                        |
| V33 | Ramification over $T=1$                                             | all odd primes $\leq 127$                                         | exact factorisation of $f_{p-1}$ and discriminant order at $T=1$ . Status: PASS.                                                                                                    |
| V34 | Necessary condition for the Morse property                          | all odd primes $\leq 127$                                         | exact Morse diagnostics and arithmetic necessary condition. Status: PASS.                                                                                                           |
| V35 | Product monodromy (finite permutation model)                        | explicit disjoint $S_3 \times S_2$ permutation action             | SymPy permutation-group order and commuting-factor check. Status: PASS. Finite model of the product-cover statement.                                                                |
| V36 | Edge count for rooted forests                                       | all rooted forests $\leq 7$ vertices                              | exhaustive exact edge counts. Status: PASS.                                                                                                                                         |
| V37 | Leaf-excess identity                                                | same exhaustive forest family                                     | exact outdegree summation. Status: PASS.                                                                                                                                            |
| V38 | Leaf lower bounds for abstract forests                              | same exhaustive forest family                                     | exact leaf/root inequalities. Status: PASS.                                                                                                                                         |
| V39 | Size bounds for complete fronts                                     | every complete front in same family                               | exhaustive front enumeration. Status: PASS.                                                                                                                                         |
| V40 | Bounds for number of complete fronts                                | same exhaustive forest family                                     | exact complete-front counts. Status: PASS.                                                                                                                                          |
| V41 | Cut-conservation identity                                           | descendant-leaf and transition-mass conserved weights             | exact rational cut sums. Status: PASS.                                                                                                                                              |
| V42 | Descendant-leaf conservation                                        | every front in same family                                        | exact descendant-leaf weights. Status: PASS.                                                                                                                                        |
| V43 | Kraft-type equality                                                 | every front in same family                                        | exact rational transition probabilities. Status: PASS.                                                                                                                              |
| V44 | Componentwise front-count lower bound                               | same exhaustive forest family                                     | componentwise binary-choice lower bound. Status: PASS.                                                                                                                              |
| V45 | Uniform random-front bounds                                         | same exhaustive forest family                                     | exact means and NumPy population variances. Status: PASS.                                                                                                                           |
| V46 | Exact Pratt leaf identity                                           | $1 \leq n \leq 500$                                               | exact arithmetic identity. Status: PASS.                                                                                                                                            |

| ID  | Result                                  | Finite scope                                         | Method and outcome                                                    |
|-----|-----------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------|
| V47 | Pratt leaf lower bounds                 | $1 \leq n \leq 500$                                  | exact derived inequalities. Status: PASS.                             |
| V48 | Size of a complete Pratt front          | all fronts for $1 \leq n \leq 60$                    | exact exhaustive front-size bounds. Status: PASS.                     |
| V49 | Number of complete Pratt fronts         | $1 \leq n \leq 60$                                   | exact front-count inequalities. Status: PASS.                         |
| V50 | Weighted conservation for Pratt fronts  | all fronts for $1 \leq n \leq 60$                    | exact root-block weighted sums. Status: PASS.                         |
| V51 | Weighted homogeneity                    | $1 \leq n \leq 60$                                   | monomial weighted degrees checked exactly. Status: PASS.              |
| V52 | Additional front-count lower bound      | $1 \leq n \leq 60$                                   | exact front-count lower bound. Status: PASS.                          |
| V53 | Range of terminal count                 | all fronts for $1 \leq n \leq 60$                    | exact terminal-count range. Status: PASS.                             |
| V54 | Polynomial comparison inequalities      | $n \leq 60$ , $x$ in $\{1/3, 2/3, 1, 2, 3\}$         | exact rational polynomial comparisons. Status: PASS.                  |
| V55 | Bounds for master polynomial            | $n \leq 60$ , two rational variable boxes            | exact master-polynomial evaluations. Status: PASS.                    |
| V56 | Uniform terminal-count moments          | all fronts for $n \leq 60$                           | exact means and NumPy population variances. Status: PASS.             |
| V57 | Front-count bounds from evaluation at 2 | $1 \leq n \leq 60$                                   | exact rational bounds using $f_n(2)=n$ . Status: PASS.                |
| V58 | Degree and leading coefficient of $f_n$ | $1 \leq n \leq 60$                                   | SymPy degree and leading coefficient. Status: PASS.                   |
| V59 | Sharp logarithmic leaf bounds           | $1 \leq n \leq 60$ ; equality families $k \leq 12$   | exact leaf counts and floating logarithmic comparisons. Status: PASS. |
| V60 | Multiplicative label-product inequality | all fronts for $n \leq 60$                           | exact integer label products. Status: PASS.                           |
| V61 | Elementary product inequality           | all tuples of length $\leq 7$ with entries 2..7      | exhaustive integer products. Status: PASS.                            |
| V62 | Total number of vertices                | $2 \leq n \leq 500$                                  | exact recursive vertex counts. Status: PASS.                          |
| V63 | Height bound                            | $2 \leq n \leq 500$                                  | exact recursive heights and numerical bounds. Status: PASS.           |
| V64 | Column inequalities for C               | all primes $\leq 127$                                | exact C-column arithmetic and inequalities. Status: PASS.             |
| V65 | Field transformation                    | prime truncation $\leq 127$ ; supported $n \leq 500$ | exact rational linear algebra. Status: PASS.                          |
| V66 | Multivariate Pratt reconstruction       | 31-prime symbolic field; supported $n \leq 60$       | exact rational-function cancellation in SymPy. Status: PASS.          |
| V67 | Diagonal local increments               | $1 \leq n \leq 100$                                  | exact diagonal rational-function reconstruction. Status: PASS.        |

| ID  | Result                                               | Finite scope                                | Method and outcome                                                                                            |
|-----|------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| V68 | Normalisation of the exponential-family distribution | n in {6,12,19,30,42,60}                     | explicit positive weights and normalisation. Status: PASS.                                                    |
| V69 | First and second derivative identities               | same six forests at nonzero fields          | finite differences versus exact weighted moments. Status: PASS. Maximum numerical residual: 2.056e-11.        |
| V70 | Positive semidefiniteness of the Hessian             | same six covariance Hessians                | NumPy symmetric eigenvalues. Status: PASS.                                                                    |
| V71 | Criterion for positive definiteness                  | same six covariance Hessians                | rank comparison with affine span of sufficient statistics. Status: PASS.                                      |
| V72 | Forced null directions                               | same six forests and all active root blocks | matrix-vector null-direction norms. Status: PASS. Maximum numerical residual: 2.056e-11.                      |
| V73 | Positive semidefiniteness of C-derived Gram matrices | prime truncation $\leq 127$                 | exact Gram factorisations, Sylvester minors, and NumPy spectra. Status: PASS. min eigenvalues 0.0833, 0.00548 |

## B Reproducing the computations

The source archive contains

- `verify_root_label_fields.py`, the executable verification suite;
- `verification_results.json`, the machine-readable full report;
- `verification_results.csv`, a spreadsheet-friendly summary;
- `README_verification.md`, the command and scope description.

The suite is run from the source directory by

```
python verify_root_label_fields.py --outdir verification_output
```

It terminates with the line

```
PASS: 73 / 73 finite checks
```

when the recorded verification succeeds.

## C References and source status

### References

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- [5] O. Leka, “Are most prime numbers symmetric?,” MathOverflow question 484349, posted December 18, 2024, edited December 25, 2024, [MathOverflow 484349](#).
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